

Adaptive thermal comfort model based on field studies in five climate zones across India

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ABSTRACT

Indian residences are vulnerable to heat-driven discomfort amid the mounting prevalence of weather extremes, residential design and construction practices, and densifying urban landscapes. Therefore, it is vital to understand the thermal comfort characteristics of nationwide residences. This study proposes an adaptive thermal comfort model based on yearlong field surveys in eight cities located across five climate zones of India – the India Model for Adaptive Comfort - Residential (IMAC-R). The model prescribes the operative temperature bands for 80% and 90% thermal acceptability in correlation with the outdoor reference temperature, applicable to mixed-mode (MM) and naturally ventilated (NV) residences.

More than 80% of the Indian residential occupants experienced a neutral thermal sensation in the indoor operative range of 16.3–35 °C in response to a 5.5–33 °C variation in the 30-day outdoor running mean temperature. Comparing the proposed model with the PMV model revealed that the latter underpredicts the thermal adaptivity of Indian occupants. The model was also compared against its predecessor – India Model for Adaptive Comfort for Commercial Buildings (IMAC MM and NV), along with relevant global and regional thermal comfort models. On average, the neutral temperature prescribed by IMAC-R was warmer than the temperatures prescribed by IMAC MM and NV by 2.9 °C and 2.1 °C, respectively; it was also warmer than the temperature prescribed by the recent ASHRAE-55 and EN 16798-1 models by 2 °C and 0.3 °C, respectively. IMAC-R reserves the prospect of addressing the thermal comfort needs of the national population while paving the way for long-term energy savings and climate action.

1. Introduction

The warming global climates have been adversely affecting the health and well-being of the global population [1–3]. The climatic marring is magnified for the densely populated regions of the tropics due to the increased frequency and intensity of heatwaves, urban heat islands, and unhealthy air quality [4–6]. In the near future, with denser cities and warmer outdoors, a high degree of dependence on active modes of

cooling will be imminent for the sustenance of the tropical population [7]. This is further confirmed by the projections of a significant increase in the global and regional cooling energy demand in the coming decades [8–10].

In the context of the rising global demand for active cooling, India is poised to emerge as the most prominent contributor to cooling-driven emissions among the major global economies. The Indian residential sector housed more than 18% of the global population in over 250 million residences in 2011 while continuing to experience a low

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Abbreviations*Term Unit Definition*

AAV	Vote Air Movement Acceptance	PPD	% Predicted Percentage Dissatisfied
AC	Air Conditioned	RAC	Residential Air Conditioner
AMV	Vote Actual Mean Vote	RH	% Indoor Relative Humidity
APV	Vote Air Movement Preference	RH _{out}	% Outdoor Relative Humidity
ASHRAE	American Society of Heating, Refrigerating, and Air Conditioning Engineers	RH _{out-month}	% Monthly Average RH _{out}
CDD	Days Cooling Degree Days	TAV	Vote Thermal Acceptance Vote
CLO	CLO Clothing Insulation	T _a	°C Air Temperature
GC	Griffith's Constant	T _{neut}	°C Neutral Temperature
HVAC	Heating Ventilation and Air Conditioning	T _{op}	°C Indoor Operative Temperature
IMAC	India Model for Adaptive Comfort - Commercial	T _{out}	°C Outdoor Air Temperature
IMAC-R	India Model for Adaptive Comfort - Residential	T _{out-rep}	°C Outdoor Representative Temperature
ISD	Integrated Surface Database	T _{out-month}	°C Monthly Average T _{out}
LL	Lower Limit	T _{out-30DRM}	°C 30-day running mean T _{out}
m	Variable Slope of line ($y = mx + c$)	T _{out-30DRM-W}	°C Weighted T _{out-30DRM}
MET	MET Metabolic Rate	T _{out-7DRM}	°C 7-day running mean T _{out}
MM	Mixed-Mode Ventilation	T _{out-7DRM-W}	°C Weighted T _{out-7DRM}
MRT	°C Mean Radiant Temperature	TPV	Vote Thermal Preference Vote
NOAA	National Oceanic and Atmospheric Administration	TSV	Vote Thermal Sensation Vote
NV	Natural Ventilation	UL	Upper Limit
OCV	Vote Overall Comfort Vote	UTCI	°C Universal Thermal Climate Index
OLS	Ordinary Least Squares Regression	V _a	m/s Indoor Air Velocity
PMV	Vote Predicted Mean Vote	V _{a-out}	m/s Outdoor Air Velocity
		WLS	Weighted Least Squares Regression
		σ_{err}^2	Error variance of T _{op}
		σ_{Top}^2	Variance of T _{op}

penetration of energy-intensive active cooling systems [11]. For instance, less than 20% of the residences in the highly urbanized National Capital Region [12,13].

For further perspective, India experiences 28% of the global share of population-weighted annual Cooling Degree Days (CDD) and experiences less than 7% RAC penetration in buildings [7,14]; in comparison, China occupies the second-highest share of population-weighted annual CDD at 10% and experiences a 60% RAC penetration [15]. With the rapidly developing urban infrastructure and growing economy, the market penetration of RACs is projected to significantly increase in India in the following decade, particularly in the cities experiencing high economic development and urbanization [14,16,17]. In the light of the role of RACs in ensuring widespread access to thermal comfort and their consequent carbon footprint, it is crucial to accurately ascertain and model the thermal inclinations of the Indian population; this helps estimate the national cooling demand while ensuring occupant well-being.

Several studies have addressed thermal comfort in India from varied thematic lenses, such as building form, building function, regional climate, and more; however, most of these studies are limited by their geographical and temporal scopes [18–23]. In 2016, Manu et al. proposed a comprehensive thermal comfort model for India based on field studies conducted in commercial buildings across multiple Indian climate zones [24]. The study was widely accepted by the national stakeholders and policymakers, and standardized as a part of the National Building Code, 2016 [25]. The present study proposes a thermal comfort model for Indian residences following a framework consistent with Manu et al.

The present study addresses the aforementioned knowledge gap through a yearlong field study of Indian residences towards formulating the India Model for Adaptive Comfort – Residential (IMAC-R). The model is based on the Right-Now-Right-Here surveys conducted across the eight cities located in the five representative climate zones of India, as shown in Fig. 1. This climate classification system is referred from the nationally-applicable code for energy conservation in buildings – the Energy Conservation Building Code, and the National Building Code [25,26].

This study provides a contextualized insight into IMAC-R through a review of relevant national and international studies, a description of occupant survey and data analysis methodology, and a comparative discussion of the research findings. The study is divided into the following sections:

- The Literature Review section delves into the basis of thermal adaptation, the need for adaptive comfort models, and the global efforts for addressing adaptive comfort. The section also helps identify the gaps based on the studies addressing adaptive comfort in Indian residences.
- The Methodology section describes the approach followed for data collection and quality assurance. The section also details the step-by-step approach followed for the mathematical analysis of the survey data.
- The Results section provides tabulations and visualizations of the survey findings and the statistical analysis. The section delves into the results on outdoor weather conditions, indoor environment conditions, occupant characteristics, subjective responses, and the thermal comfort model.
- The Discussion section expands upon the implications of the thermal comfort model. The section discusses the model vis-à-vis relevant global and national thermal comfort models and the scope of future research.
- The Conclusion section summarizes the key research findings and emphasizes the importance of policy-level interventions in line with the national vision of ‘thermal comfort for all.’

Furthermore, the study includes ten Annexures that provide additional insights to promote research replicability by offering a deeper context into the occupant survey, nature of indoor and outdoor data, and analysis approach. The following section, as described above, reviews the literature on the adaptive theory and its applications in India.

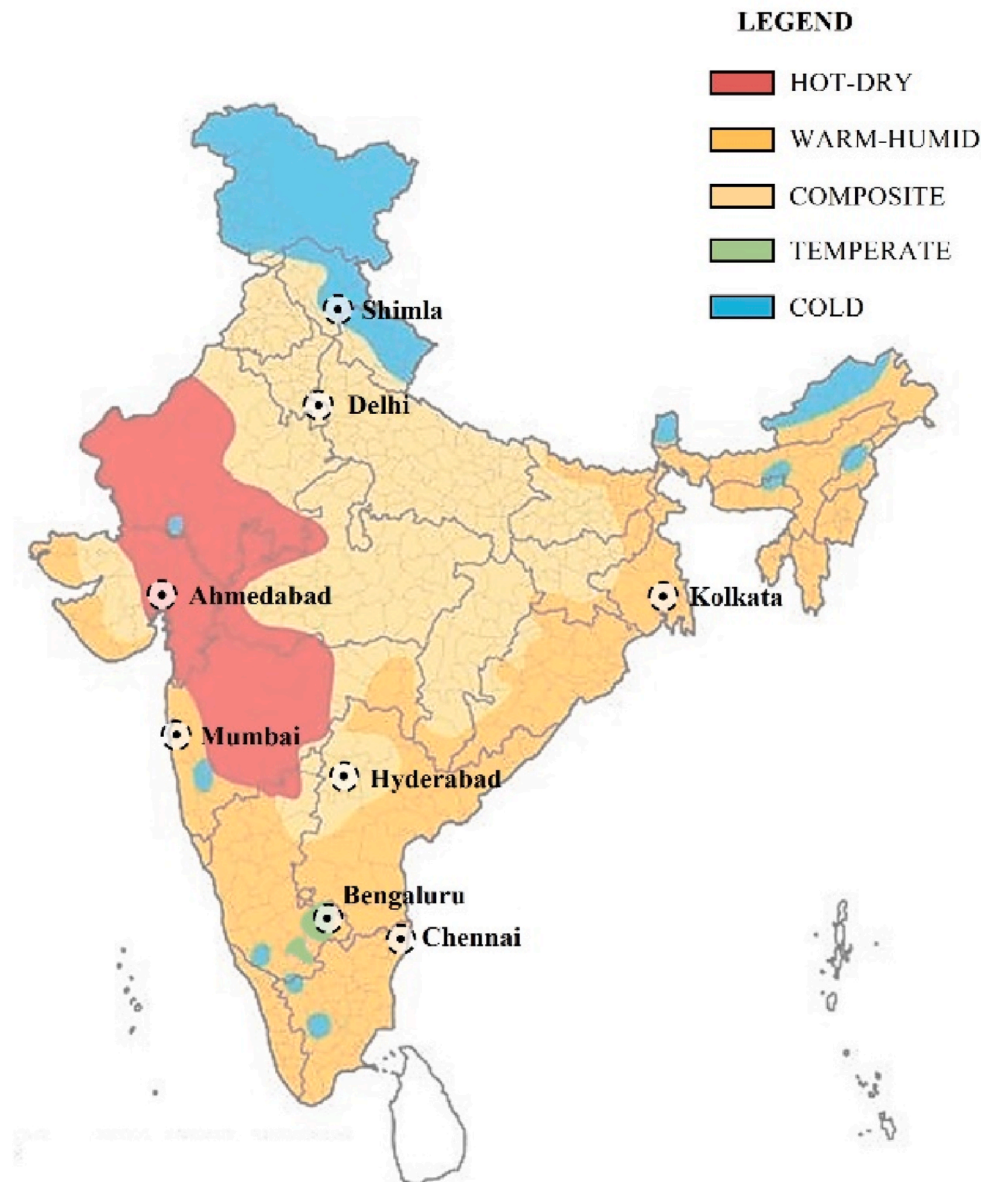


Fig. 1. The eight surveyed cities in the five Indian climate zones.

2. Literature review

The projected increase in the adoption of RACs to address the cooling demand of the Indian residential population can be better managed by appropriate building design and operation, accounting for the occupants' thermal adaptation. Adaptive comfort models capture this behavioral nuance of the occupants, and their widespread adoption can help improve the national energy efficiency scenario. Adopting thermal comfort models can also aid in ensuring climate resilience for a significant fraction of building envelopes while maintaining widespread access to thermal comfort.

2.1. Adaptive comfort models

The human thermal perception is influenced by three key factors: behavioural adjustment, physiological acclimatisation, and psychological expectation [27]. Behavioural adjustments account for (a) occupant-level changes to clothing insulation (CLO) and metabolic rate (MET), and (b) envelope-level interventions such as window and fan operation [28,29]. Physiological acclimatisation – also known as

thermal adaptation – accounts for the bodily responses to the prolonged prevalence of characteristic ambient thermal conditions [30]. Lastly, the psychological expectations account for the occupants' subjective preference for a warmer or cooler thermal sensation. An adaptive comfort model combines these three factors to prescribe comfortable indoor temperature with respect to a representative outdoor temperature ($T_{out-rep}$).

In contrast to the static setpoint-based conditioning of the indoor environment, adaptive comfort model-based conditioning prescribes the indoor setpoint for the acceptability of up to 90% of the occupants at granular timescales. The often-significant offset between the static and adaptive setpoints provides the opportunity for energy savings and maintenance of comfortable indoor environments [31]. The energy savings stem from the decreased load on the active cooling systems due to relatively relaxed thermal setpoints. Studies have demonstrated that precedence to adaptive setpoints over static setpoints can help reduce the average cooling energy consumption by ~30% and peak electrical energy demand by ~70% in warm climates [31,32].

While multiple studies established the significance of adaptive comfort since the early 1970s [33–35], it was first formalised into a

thermal comfort Standard in the form of the ASHRAE Standard 55 in 2004 [36]. The Standard included the model proposed by de Dear and Brager based on thermal comfort data from diverse global climates [37]. Subsequently, multiple adaptive models have been introduced globally – Table 1 shows the current international Standards and Codes with their adaptive regression models and region of suggested applicability based on the data source.

The adaptive model – ASHRAE-55, 2020, accounts for global thermal comfort data representing all major climatic regions, including the tropics [41]. However, a localised adaptive model comprising country/region-specific data points is more representative of the occupants' thermal preferences. This difference is reflected in the regression slopes (representative of the thermal adaptation) for ASHRAE-55 and IMAC models [24,38]. It is important to note that the National Building Code of India does not define the applicable building typology due to the limitations of its erstwhile scope; however, the dataset of its inherent adaptive model (IMAC) comes from the monitoring of commercial buildings [24]. Building typology is likely to influence thermal adaptation; therefore, it is pertinent to devise a thermal comfort model based on the adaptive thermal comfort theory relevant to Indian residences.

2.2. Adaptive comfort studies in India

Given the mounting need for thermally comfortable buildings in India due to increased urban heat stress, several studies have examined thermal comfort in India. Some of these studies have explored the effect of occupant-level interventions, envelope-level interventions, and building typology on thermal comfort [18–20,42–44]; some studies have devised region-specific adaptive comfort models using an occupant survey-based approach, similar to the present study [20–23]. Over-archingly, these studies reveal that the Indian population subsets were more thermally adaptable relative to the global population. In 2016, Manu et al. published a thermal comfort model based on the adaptive

theory, with proposed applicability across the five Indian climate zones; this model was termed as India Model for Adaptive Comfort (IMAC), and it was based on yearlong surveys across commercial buildings [24]. Table 2 summarizes the recent adaptive comfort models for India, their climatic applicability, scope, and key regression equation.

The IMAC study by Manu et al. was based on a yearlong survey of 6330 responses from 16 buildings across the five Indian climate zones. The study prescribed adaptive comfort models for commercial buildings operating using Mixed-Mode Ventilation (MM) and Natural Ventilation (NV). The analysis was performed using Griffith's Method. The Griffith's Constants (GC) for MM and NV buildings were calculated as 0.16 and 0.15, respectively, and used to derive the thermal neutralities. The occupant responses were binned as per 'Building + Season', and the thermal acceptability bands were derived using Weighted Least Squares (WLS) Regression.

However, there remains a need for thermal comfort studies that can help formulate a residential thermal comfort model; these studies should incorporate temporally consistent surveys and validated methodology in the parlance of Indian residences. Adaptive comfort in residences is based on varied occupant behaviours, indoor environments, and envelope characteristics. While select national studies have adequately addressed residential thermal comfort for specific regions of India, there remains a lack of a nationwide model representing yearlong adaptive thermal comfort in residences across the five Indian climate zones [45].

2.3. Summary

The existing literature on adaptive comfort in India establishes the need for a nationwide understanding of residential thermal comfort to strengthen the recommendations from the National Building Code. This study attempts to bridge this knowledge gap by proposing the India Model for Adaptive Comfort – Residential (IMAC-R). It builds upon the field survey and data analysis framework established by IMAC and

Table 1

Current international Standards and Codes with adaptive comfort models and their region of geographic applicability; here, UL: Upper Limit, and LL: Lower Limit.

No.	Standard or Code	Region of Applicability	Relevant Adaptive Model (90% Acceptability)	Definition of $T_{out-rep}$
1	ANSI/ASHRAE - 55, 2020 [38]	World	NV Operation UL, $T_{comf} = 0.31(T_{out-rep}) + 20.2$ LL, $T_{comf} = 0.31(T_{out-rep}) + 15.5$	Prevailing mean outdoor temperature (7-day to 30-day horizon) T_{out} Applicability Range: 10–34 °C
2	Chinese GB/T 50785, 2012 [39]	China	NV Operation UL, $T_{comf} = 0.77(T_{out-rep}) + 9.34$ LL, $T_{comf} = 0.87(T_{out-rep}) + 0.31$	Running mean outdoor temperature (7-day horizon). T_{out} Applicability Range for, UL: 11.2–24.2 °C LL: 21–32.5 °C
3	ISSO 74, 2014 [40]	Europe	NV Operation <i>New Buildings and Buildings for Vulnerable Occupants</i> UL, $T_{comf} = 0.33(T_{out-rep}) + 20.8$ LL, $T_{comf} = 0.2(T_{out-rep}) + 18$ <i>Existing Buildings</i> UL, $T_{comf} = 0.33(T_{out-rep}) + 21.8$ LL, $T_{comf} = 0.2(T_{out-rep}) + 17$	Running mean external temperature (approximate equation with 7-day horizon) T_{out} Applicability Range: 10–25 °C
4	National Building Code of India, 2016 [25]	India	NV Operation UL, $T_{comf} = 0.54(T_{out-rep}) + 15.23$ LL, $T_{comf} = 0.54(T_{out-rep}) + 10.43$ MM Operation UL, $T_{comf} = 0.28(T_{out-rep}) + 21.37$ LL, $T_{comf} = 0.28(T_{out-rep}) + 14.37$	30-day outdoor running mean temperature T_{out} Applicability Range for, NV: 12.5–31 °C MM: 13–38.5 °C
5	EN 16798-1, 2019 [40]	Europe	NV Operation <i>Buildings for Vulnerable Occupants</i> UL, $T_{comf} = 0.33(T_{out}) + 20.8$ LL, $T_{comf} = 0.33(T_{out}) + 16.8$ <i>New Buildings</i> UL, $T_{comf} = 0.33(T_{out-rep}) + 21.8$ LL, $T_{comf} = 0.33(T_{out-rep}) + 15.8$ <i>Existing Buildings</i> UL, $T_{comf} = 0.33(T_{out-rep}) + 22.8$ LL, $T_{comf} = 0.33(T_{out-rep}) + 14.8$	Running mean external temperature (approximate equation with 7-day horizon) T_{out} Applicability Range: 10–30 °C

Table 2
Comfort models for India and its regions.

No.	Study	Climate(s)	Scope and Applicability	Thermal Neutrality
1	Indraganti et al., 2014 [21]	Warm-Humid	NV Commercial Buildings	$T_{neut} = 0.26(T_{out-30DRM-W}) + 21.40$
2	Indraganti et al., 2014 [21]	Warm-Humid	AC Commercial Buildings	$T_{neut} = 0.15(T_{out-30DRM-W}) + 22.10$
3	(IMAC, NV)	All	NV Commercial Buildings	$T_{neut} = 0.54(T_{out-30DRM}) + 12.83$
4	Manu et al., 2016 [24]	All	MM Commercial Buildings	$T_{neut} = 0.28(T_{out-30DRM}) + 17.87$
	(IMAC, MM)			
5	Manu et al., 2016 [24]			
6	Thapa and Indraganti, 2020 [23]	Warm-Humid	All NV Buildings	$T_{neut} = 0.45(T_{out-30DRM-W}) + 16.45$
7	Thapa and Indraganti, 2020 [23]	Cold	All NV Buildings	$T_{neut} = 0.55(T_{out-30DRM-W}) + 10.80$
8	Thapa et al., 2018 [20]	Cold	NV Buildings	$T_{neut} = 0.69(T_{out-30DRM-W}) + 8.91$
	Thapa, 2020 [22]	Cold	NV Residences	$T_{neut} = 0.60(T_{out-30DRM-W}) + 10.40$

$T_{out-30DRM}$ is the 30-day running mean of T_{out} , and $T_{out-30DRM-W}$ is weighted $T_{out-30DRM}$

ASHRAE, with relevant validated modifications. The modelled findings are compared against the findings from Manu et al. to assess the difference in thermal adaptation and expectations of occupants in commercial buildings and residences in India. The proposed model is also compared against two international models (ASHRAE-55 and EN-16798-1) and regional models for warm and cold climate zones of India. The following section provides an insight into the data collection, quality assurance, and analysis approaches followed in the study.

3. Methodology

The data collection methodology draws from the ASHRAE RP-884 project and follows the CLASS-II measurement protocols; the analysis methodology relies on the Griffith's Method used for IMAC [24,46]. The following subsections detail the inherent methodologies for data collection, quality assurance, and analysis.

3.1. Data collection

3.1.1. Identification of cities and residences

The study of adaptive comfort in Indian residences commenced with identifying the nationwide data collection methodology. It was defined while considering the climatic representativeness of the Indian cities and the feasibility of year-long comfort monitoring. Eight cities from five Indian climate zones (given in Table 3) were identified with a context of the representative residential typologies. The geographical spread of Indian climate zones is illustrated in Fig. 1.

The survey monitored thermal conditions across 294 residences representing the eight typologies described in Table 4. Half of the surveyed residences were NV, i.e., the residences utilized fans and windows to maintain thermal comfort across all the spaces. The other half of the surveyed residences were MM, i.e., the residences utilized active modes of heating/cooling in addition to fans, in at least one of the spaces. While the residences are classified as NV and MM, the individual responses are classified as NV or Air Conditioned (AC) based on the operation of a space during the thermal comfort survey [24]. If the surveyed space utilized fans, the corresponding responses were binned as NV; whereas, if the surveyed space utilized any additional active mode of

Table 3
The cities identified for thermal comfort surveys and their relevant details.

Climate Zone	Cities	City ID	Level of Development [47]
Hot-Dry	Ahmedabad	AMD	Tier 1
Warm-Humid	Chennai, Mumbai, Kolkata	CHN, MUM, KOL	Tier 1 each
Composite	Delhi, Hyderabad	DEL, HYD	Tier 1 each
Temperate	Bengaluru	BLR	Tier 1
Cold	Shimla	SHM	Tier 2

heating/cooling, the corresponding responses were binned as AC. The NV and AC Bins are tabulated in Annexure 7.8.

3.1.2. Survey and monitoring details

The surveys recorded 'background data' for site-specific parameters describing the residence envelope and the socio-economic context, and 'recurrent data' for (a) local weather conditions, (b) indoor environment parameters describing the thermal and air movement characteristics, and (c) Right-Now-Right-Here (RNRH) surveys quantifying occupants' response to varied indoor environments on psychological voting scales, given in Table 5. The verbal anchors of the voting scales were translated into seven local languages – Hindi (DEL, SHM, and others), Gujarati (AHD), Marathi (MUM), Tamil (CHN), Telugu (HYD), Kannada (BLR), and Bangla (KOL); the translations are included as Annexure 7.1.

The data was collected using online forms by the (city-specific) survey team and verified by the central team, located in Ahmedabad, on a day-to-day basis. The three forms used for data collection are briefly described in Annexure 7.2. The survey and monitoring recorded over 50 background and recurring parameters; these parameters are tabulated and defined in Annexure 7.3. The yearlong field studies adhered to the following structure:

- The residences were identified, and the occupants were briefed about the nature and duration of the study.
- After the introductory site visit, the occupants were approached at least two days before the thermal comfort survey to ensure their presence at the residence. Thereafter, the survey visits were planned once per month, based on the occupants' availability.
- During the subsequent surveys, the eight independent surveying teams – situated in the eight cities – followed a standard operating procedure based on a centrally enforced checklist. This checklist heavily emphasised the practices to limit the spread of the SARS-CoV-2 virus and ensure data validity.

Table 4
Description of surveyed residential typologies.

Typology	Description	Responses
Affordable Housing	Apartment Units with 1 Room + Kitchen (1 RK or 1 BHK)	240
Apartments	Apartment Units with 2 Bedrooms + Hall + Kitchen (2 BHK)	1053
Bungalow	Individual House	337
Duplex	Apartment Units with two floors	191
Penthouse	Apartment Units with two floors, including the top floor	15
Rowhouse	Individual House sharing a common wall with adjacent building	126
Tenement	Individual House on Ground + 1 (G+1) floor	186
Traditional House	Individual House with a traditional/vernacular building material	31

Table 5

Psychological voting scales of sensation, preference, and acceptance of perceived temperature and air movement, along with overall thermal comfort [36].

Vote	Thermal Sensation Vote (TSV)	Thermal Preference Vote (TPV)	Thermal Acceptance Vote (TAV)	Air-Movement Acceptance Vote (AAV)	Air-Movement Preference Vote (APV)	Overall Comfort Vote (OCV)
−3	Cold	-	-	-	-	Very Uncomfortable
−2	Cool	Want Cooler	Very Unacceptable	Very Unacceptable	-	Uncomfortable
−1	Slightly Cool	Want Slightly Cooler	Unacceptable	Unacceptable	Want More Air Movement	Slightly Uncomfortable
0	Neutral	No Change	-	-	No Change	-
1	Slightly Warm	Want Slightly Warmer	Acceptable	Acceptable	Want Less Air Movement	Slightly Comfortable
2	Warm	Want Warmer	Very Acceptable	Very Acceptable	-	Comfortable
3	Hot	-	-	-	-	Very Comfortable

- The surveys were conducted in the living room or the bedroom of the residence. They were conducted between 09:00 and 21:00 h, with a majority falling between 18:00 and 19:30 h.
- The occupants were surveyed for the parameters described in Annexure 7.3. Simultaneously, the indoor environment was monitored while ensuring the stabilisation of the instruments described in the subsequent sub-section.
- The survey team also accounted for the operational status of active modes of ventilation, heating, cooling, as well as window operation.

Additionally, the time-series data on local weather conditions were collected from three sources: (a) Global Hourly Integrated Surface Database (ISD) by NOAA [48], (b) locally installed sensors for outdoor temperature measurement, and (c) daily peak data from local newspapers. The data from (a) was used for further analysis, while the data from (b) and (c) were used for data consistency and quality checks. The high-resolution ISD data provided hourly values for over 30 outdoor parameters for the eight surveyed cities – the parameters are briefly described in Annexure 7.4.

3.1.3. Instrumentation for indoor measurements

The indoor environment parameters of Air Temperature (T_a), Relative Humidity (RH), and Air Velocity (V_a) were measured during the surveys using TSI Velocicalc Air Velocity Meter 9545-A, and the Globe Temperature (T_g) was measured using Extech HT 200 Heat Stress Meter, in line with ASHRAE Class 2 specifications [37]. The respective ranges and accuracies of the indoor measurements are given in Table 6.

3.2. Quality assurance

The raw data was processed through the three steps of (a) indices calculation, (b) data validation, and (c) data cleaning to yield the data for further analysis. The three steps and their components are illustrated in Fig. 2 and explained below.

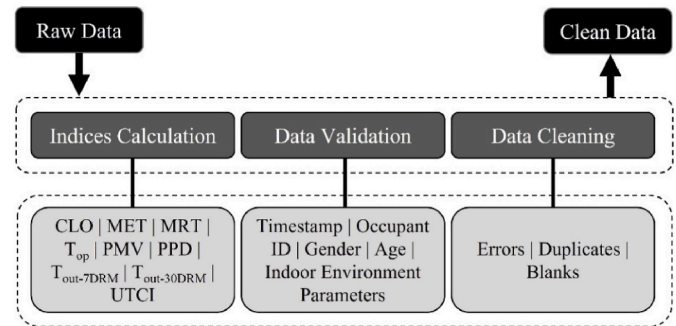
3.2.1. Indices calculation

The indices of Clothing Insulation (CLO), Metabolic Rate (MET), Mean Radiant Temperature (MRT), Operative Temperature (T_{op}), Predicted Mean Vote (PMV), Predicted Percentage Dissatisfied (PPD), 7-day

Table 6

Instrumentation details.

Device	Indoor Parameters	Range	Accuracy	Resolution
TSI Velocicalc Air Velocity Meter 9545-A	Air Temperature (°C)	−10 to 60 °C	±0.3 °C	0.1 °C
	Relative Humidity (%)	5–95%	±3%	0.1%
	Air Velocity (m/s)	0–50 m/s	±3% or ±0.015 m/s	0.01 m/s
Extech HT 200 Heat Stress Meter	Globe Temperature (°C)	0–80 °C	±0.6 °C	0.1 °C

**Fig. 2.** Quality Assurance framework.

and 30-day Outdoor Running Mean Temperatures ($T_{out-7DRM}$ and $T_{out-30DRM}$), and Universal Thermal Climate Index (UTCI) were calculated using a combination of survey data and ISD weather data.

CLO was calculated by converting (checkbox-based data to numeric data) and combining the clothing insulation values of individual clothing items worn by the occupants during the survey. Similarly, MET was calculated based on the activity levels of the occupants before and during the survey. The corresponding reference clothing insulation values and metabolic rates were referred from the ASHRAE database [38]. MRT, T_{op} , PMV, and PPD were calculated using the respective ‘calcTroin’, ‘calcPMV’, and ‘calcPPD’ functions of the R comf package [49]; these functions follow the methodology prescribed in ISO 7726:2001 [50]. Similarly, $T_{out-7DRM}$ and $T_{out-30DRM}$ were also calculated using user-defined functions in R. The outdoor heat stress was assessed as UTCI; it was calculated using BioKlima – a tool for calculating bioclimatic and thermophysiological parameters [51].

The MRT and T_{op} calculation required four input parameters – Air Temperature (T_a), Air Velocity (V_a), Globe Temperature (T_g), and MET from the field surveys. The PMV and PPD calculations required six input indoor parameters – T_a , RH, V_a , MRT, MET, and CLO, procured from the field surveys. The outdoor representative temperature ($T_{out-rep}$) is defined as two metrics: $T_{out-7DRM}$ and $T_{out-30DRM}$; their calculation requires Outdoor Air Temperature (T_{out}), which is referred from the ISD weather data. Similarly, the UTCI calculation requires the input of T_{out} , RH_{out}, V_{a-out} , Atmospheric Pressure, Solar Radiation (Global, Diffused, and Direct), and geographical latitude from the ISD weather data.

3.2.2. Data validation

The validity of individual data points was checked against various parameter-specific criteria; the sequential steps of data validation are given as follows:

- The online timestamp of the survey was corroborated against the actual date and time of the survey.
- The Occupant ID (a unique alphanumeric code including occupant age and gender) was checked for consistency with the recorded gender of each occupant.

- iii. The Occupants' age group was verified against the background data.
- iv. The indoor environment parameters were checked to fall within an acceptable range. These ranges varied by the climate zone/city and outdoor weather conditions. The parameter-specific ranges of the clean data are given in Annexure 7.5.

3.2.3. Data cleaning

The datapoints with erroneous, duplicate, or blank entries were either removed or modified as per the following steps:

- i. The datapoints with erroneous or duplicate entries for indoor environment parameters (T_a , V_a , RH, and T_g) were removed. Of the 2691 total responses, 409 erroneous and 53 duplicate entries were removed.
- ii. The datapoints with a significantly high (over 5 °C) difference between T_a and MRT were flagged. Out of the 2691 total responses, 50 such entries were removed.
- iii. The datapoints with semantic errors were flagged and modified individually.

The above steps were performed using a combination of R and MS-Excel methods to yield 2179 clean responses (out of 2691). The additional details on data cleaning are provided in Annexure 7.6.

3.3. Analysis approach

The 2179 clean survey responses were analyzed using two regression approaches (Linear Regression and Griffith's Method) and 12 data binning approaches (such as 'Location + Month + Mode,' 'Location + Day,' 'Location + Season,' and more) based on the volume and nature of the collected datapoints. Annexure 7.7 lists all the explored analysis approaches and the consequent regression models. Among these approaches, binning the data by 'Location + Month + Mode' and analyzing it using Griffith's Method was identified as the most appropriate approach for further analysis. This approach was selected on the basis of the volume of significant data towards the model formulation and the congruence of relevant statistical parameters (such as the Calculated Griffith's Constant) with the literature.

Griffith's Method was preferred as it allowed 100% of the 2179 data points to be utilized for formulating the final thermal comfort model. Simultaneously, the 'Location + Month + Mode' bins yielded a Griffith's Constant (GC) of 0.15, which was closest to the GC stipulated by IMAC (MM = 0.15, NV = 0.16) compared to the other binning approaches. The identified analysis approach involved the following steps:

- i. The clean data was distributed by 'Location + Month + Mode' into 82 bins; each bin included the data for a unique city (from the eight surveyed cities), a unique month (from the 12 months), and a mode of operation of surveyed space (NV or AC). The bin-wise description of data is included in Annexure 7.8.
- ii. For each of the 82 bins, the ' ΔTSV ' and ' ΔT_{op} ' were calculated using Equation (1) and Equation (2):

$$\Delta TSV = \text{Avg_TSV} - TSV \quad (1)$$

$$\Delta T_{op} = \text{Avg_}T_{op} - T_{op} \quad (2)$$

Where, TSV = Thermal Sensation Vote, Avg_TSV = Bin Average TSV, T_{op} = Operative Temperature, Avg_ T_{op} = Bin Average T_{op} .

- iii. The linear regression of ΔTSV (ordinate) and ΔT_{op} (abscissa) for 2179 datapoints was used to calculate the GC as 0.152, using Equation (3) and Equation (4):

$$m = \Delta TSV / \Delta T_{op} \quad (3)$$

$$GC = m \cdot (\sigma_{T_{op}}^2 / \sigma_{TSV}^2 - \sigma_{err}^2) \quad (4)$$

Where, m = slope of the ΔTSV vs ΔT_{op} linear regression (calculated as 0.151), $\sigma_{T_{op}}^2$ = variance of 2179 T_{op} datapoints (calculated as 2.21), and σ_{err}^2 = error variance of T_{op} (taken as 0.158 from Humphreys et al. [52]).

The GC was calculated as 0.152 – in more tangible parlance, it implies that a unit change in T_{op} results in a change in TSV by 0.152 votes in (MM and NV) Indian residences; inversely, a unit change in TSV can be induced through a 6.6 °C change in T_{op} . The GCs calculated in the IMAC study for MM and NV residences were 0.15 and 0.16, respectively. Considering the ease of policy-level implementation of IMAC across building typologies and climate zones and the physical significance of the two GCs, the IMAC study used GC = 0.16 for devising both MM and NV models for commercial buildings. Therefore, to ensure consistency and comparability of findings subject to the observed data points, the present study also uses GC = 0.16 (instead of the Calculated GC = 0.152) for further analysis.

- iv. With GC = 0.16, the neutral temperature (T_{neut}) – corresponding to the indoor operative temperature (T_{op}) at which the occupant has a neutral thermal sensation (TSV = 0) – was calculated for each of the 82 bins using Equation (5).

$$T_{neut} = \text{Avg_}T_{op} - (\text{Avg_}TSV / GC) \quad (5)$$

- v. Bin-specific values of T_{neut} (ordinate) were regressed against the corresponding bin-average values of $T_{out-30DRM}$ (abscissa) using the Weighted Least Squares (WLS) method; the weights were defined as the number of responses in the respective bins. The consequent regression line represents the thermal neutrality for all Indian residences; it represents 100% thermal acceptability.
- vi. Parallely, the 80% and 90% thermal acceptability bands, represented by ΔT_{op} , were derived using the sensitivity-slope method used for IMAC. The linear regression equations of the form ' $TSV = \text{slope} \cdot (T_{op}) + c$ ' for the 'Location + Month + Mode' bins were identified, and their statistical significance was tested as per the steps included in Annexure 7.9. The slopes (representing occupant thermal sensitivity) of the statistically significant bins were averaged using their bin sizes as weights. The Weighted Average Slope (calculated as 0.23) was used to define the thermal acceptability bands (ΔT_{op}) using Equation (6).

$$\Delta TSV = (\text{Weighted Average Slope}) \cdot \Delta T_{op} \quad (6)$$

- vii. To calculate ΔT_{op} bands of thermal acceptability, ΔTSV was taken as 0.85 and 0.5, corresponding to a neutral sensation (TSV = 0) of 80% and 90% of occupants, respectively. The band was symmetrically divided (as $\Delta T_{op}/2$) to derive the Upper and Lower Limits (UL and LL) of thermal acceptability. These 80% (± 3.60 °C) and 90% (± 2.15 °C) limits were combined with the T_{neut} derived from Equation (5) to yield the final acceptability bands.
- viii. The final model was proposed in correlation with three variables in $T_{out-7DRM}$, $T_{out-30DRM}$, and UTCI for context-specific applicability. The following section illustrates and discusses the survey and modelling results.

4. Results

Devising the proposed thermal comfort model involved the analysis of the outdoor weather conditions in correlation with the indoor environment parameters and the occupant-level characteristics from the RNRH survey responses. This section expands upon the outdoor weather conditions, first-hand survey insights, occupant details, corresponding indoor environments, and the adaptive comfort models for Indian residences.

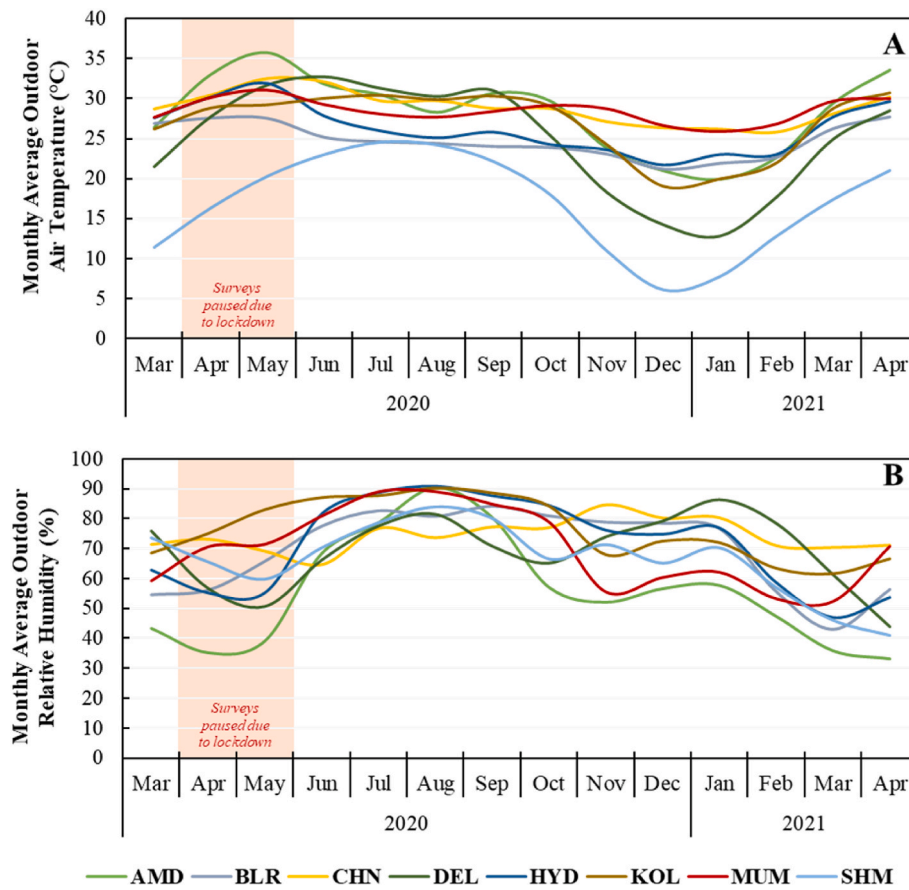


Fig. 3. Variation of monthly average outdoor (a) Air Temperature ($T_{\text{out-month}}$) and (b) Relative Humidity ($RH_{\text{out-month}}$) for the eight surveyed cities.

4.1. Outdoor weather conditions

The nationwide survey was conducted between March 2020 and April 2021, excluding April and May 2020 due to the pandemic-driven travel restrictions. Fig. 3 shows the Monthly Average Outdoor Air Temperature ($T_{\text{out-month}}$) and Relative Humidity ($RH_{\text{out-month}}$) for the eight cities during the survey period. The $T_{\text{out-month}}$ was observed to be

the highest (33.6°C) during the summer months (Mar–Jun) for the Composite Climate (DEL, HYD), while it was the lowest (6°C) during the winter months (Nov–Feb) for the Cold Climate (SHM). Similarly, the RH_{out} was the highest for the Composite Climate (90.8%) during the monsoon months (Aug–Oct) and lowest (33.2%) for Hot-Dry Climate (AMD) during the summer months.

Occupants' thermal inclinations depend on several climatic

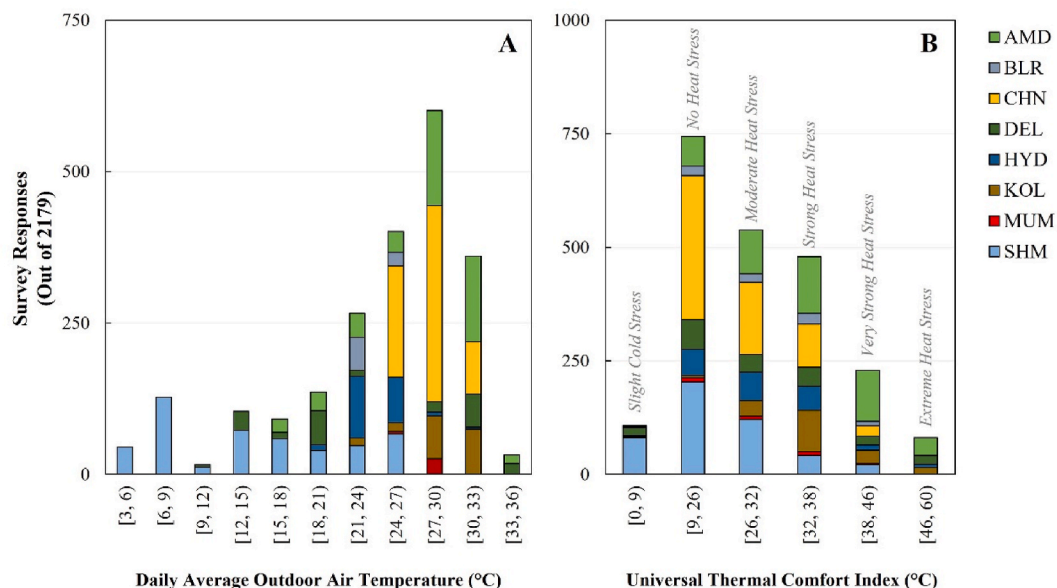


Fig. 4. City-wise distribution of (a) daily average outdoor air temperature ($T_{\text{out-day}}$) and (b) UTCI during the surveys.

parameters in addition to the outdoor air temperature. Therefore, it can be helpful to examine the variation of outdoor heat stress consequent to the T_{out} , RH_{out} , outdoor dew-point temperature, solar radiation, outdoor air velocity, and atmospheric pressure across the eight surveyed cities. Universal Thermal Climate Index (UTCI) helps quantify outdoor heat stress while accounting for these six climatic parameters. UTCI prescribes index-specific non-homogeneous bands, indicating the presence and extent of thermal stress on an occupant in response to the prevalent climatic conditions.

Fig. 4 shows the distribution of survey responses with respect to the bins of daily average outdoor air temperature ($T_{out-day}$) and UTCI. The figure shows that although certain cities indicate a high number of responses corresponding to a high $T_{out-day}$, their UTCI provides a better insight into the occupant-perceived heat stress. For instance, AMD has fewer responses with $T_{out-day} \geq 24^\circ\text{C}$ in comparison to CHN, yet AMD has a higher number of responses corresponding to ‘strong’ to ‘extreme’ heat stress. Therefore, to complement the standard comfort modelling approach (with $T_{out-rep}$ as the independent variable), it is also helpful to establish the correlation between occupants’ thermal inclinations and outdoor heat stress (with UTCI as the independent variable).

4.2. National survey of residential occupants

The year-long survey examined occupants from 294 residences resulting in 2179 valid responses. The survey accounted for over 50 background and recurring parameters addressing the residences and occupants, as described in Annexure 7.3. The preliminary insights from the survey are given below.

4.2.1. Residences and occupants

The survey accounted for residences and occupants representing the typical Indian residential sector. In the process, the survey recorded the construction age, residence type (Low-, Mid-, and High-rise), and the surveyed floor (Ground, Middle or Intermediate, and Top) – the findings are illustrated in Fig. 5. A majority of the survey responses came from New (Construction Age <25 years), Low-rise residences, with occupants residing on the Ground and Middle floors.

Along with the age and type of residence, the occupant Age-Group and Gender were also recorded. The surveys recorded a high (>1) Female/Male response ratio; the relevant demographic details are provided in Table 7.

As shown in Fig. 6, a majority of the occupants participated in the surveys once. In other words, 307 occupants were surveyed once (resulting in 307 responses), 192 of them were surveyed twice (resulting in 384 responses), 99 of them were surveyed thrice (resulting in 297 responses), and similarly for other cases, as illustrated in the figure.

The surveyed occupants were typically dressed in conventional

Table 7
Occupant demographics.

Occupant Age Group	Responses (Out of 2179)	Female/Male Ratio
19–30	732 (34%)	1.16
31–40	396 (18%)	1.30
41–50	447 (20%)	1.83
51–62	604 (28%)	1.09

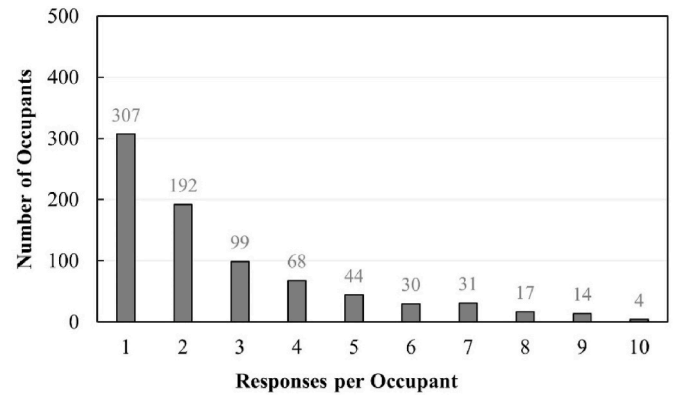


Fig. 6. Distribution of the number of responses per occupant.

Indian clothes (for females – saree/kurta/t-shirt with petticoat/pyjama/jeans and undergarments; for males – shirt/t-shirt with jeans/pants and undergarments); over 90% of the responses indicated a Clothing Insulation (CLO) value of 0.1–1.0 CLO. Additionally, the occupants were involved in light to moderate indoor activities such as lying, sitting, standing, walking, reading, and cleaning; over 90% of the responses indicated a Metabolic Rate (MET) of 0.7–1.7 MET. The spread of CLO and MET for the survey responses is illustrated in Fig. 7.

The adaptive response of the surveyed occupants can be understood by examining the variation in CLO, V_{air} , and MET with respect to binned values of operative temperature (T_{op}), as shown in Fig. 8. With a 10°C increase in T_{op} , the occupants’ adaptive response was reflected by a 0.41 CLO decrease in the clothing insulation, 0.55 m/s increase in the air velocity, and an insignificant (0.014 MET) decrease in the metabolic rate.

4.2.2. Indoor environment

Thermal comfort is defined by two personal parameters in CLO and MET, as described above, and four indoor environment parameters in Air Temperature (T_{air}), Relative Humidity (RH), Globe Temperature (T_g), and Air Velocity (V_a). Fig. 9 shows the spread of the four indoor

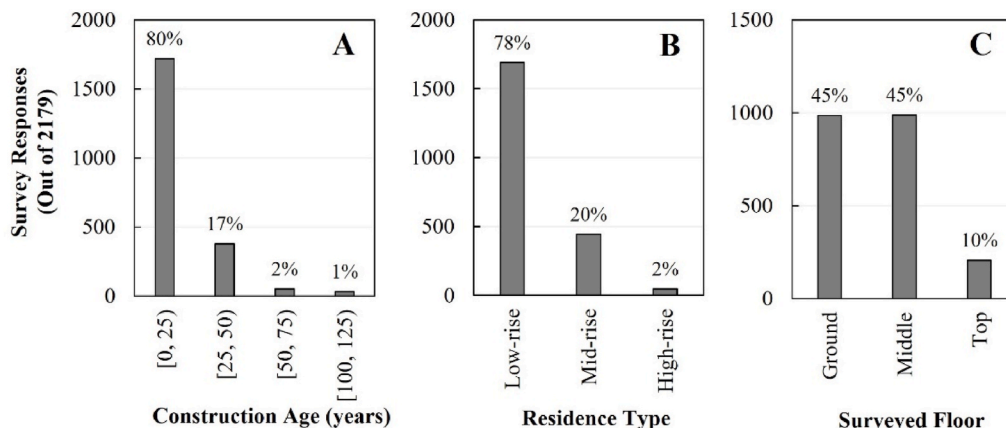


Fig. 5. The (a) construction age, (b) residence type, and (c) surveyed floor.

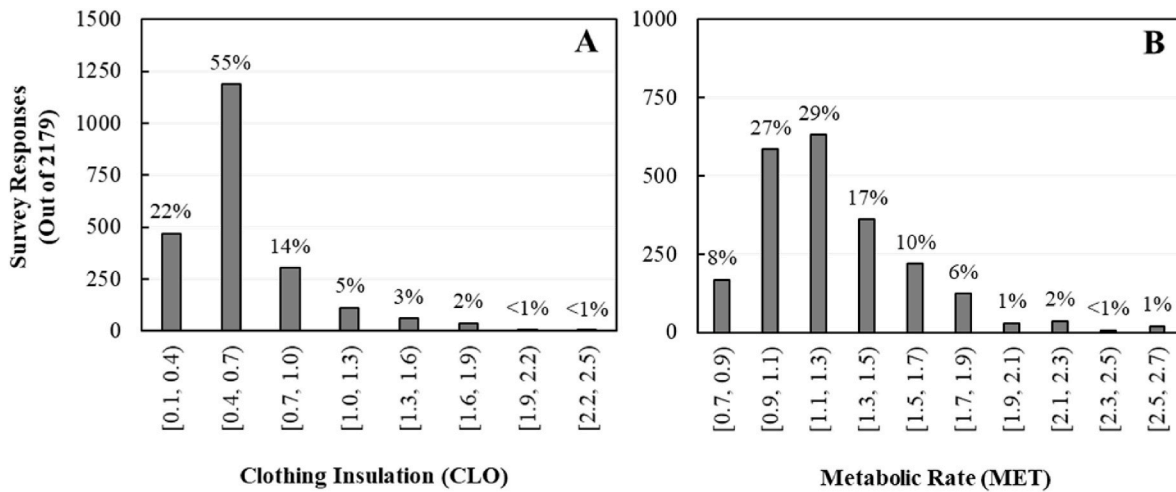


Fig. 7. Clothing insulation (CLO) and metabolic rate (MET) of the surveyed occupants.

environment parameters measured during the RNRH surveys. The respective measured ranges were observed as 7–37 °C (T_{air}), 20–95% (RH), 7–37 °C (T_g), and 0–3.7 m/s (V_a); city-specific spread of measurement ranges and survey responses can be referred to from Annexure 7.5.

With the measured indoor parameters as input, the Mean Radiant Temperature (MRT) and Operative Temperature (T_{op}) were calculated using the method described in Section 3.2.1– their spread is illustrated in Fig. 10. The measured range of MRT was observed as 6–38 °C, while the range of T_{op} was 7–37 °C.

The occupants were surveyed for psychological responses, as described in Section 3.1.2, and the results for Thermal Sensation (TSV), Acceptability (TAV), and Preference (TPV) Votes are illustrated in Fig. 11. TSV was dominantly ‘Neutral’ (48% responses), TAV was dominantly ‘Acceptable’ (75% responses), and TPV was dominantly ‘No Change’ (51% responses). These dominant psychological responses significantly contributed to the favorable response for the Overall Comfort Vote (OCV) and derivation of thermal neutralities.

The above-described indoor and outdoor environment parameters and surveyed subjective responses (for Overall Thermal Comfort, Thermal Sensation, Thermal Acceptance, Thermal Preference) were used to propose the thermal comfort model for Indian residences.

4.3. Deriving thermal comfort model

4.3.1. Comparing observed and predicted sensation

The basis of modelling thermal comfort for Indian residences lies in the lack of applicability of existing models – the ‘static-PMV’ model has been previously shown to underpredict the thermal adaptation of Indian occupants [24]. In order to highlight this deviation, the ‘Observed’ Thermal Sensation Votes (TSV) were compared against the ‘Calculated’ Predicted Mean Votes (PMV). TSV was taken from the RNRH surveys and is synonymously referred to as Actual Mean Vote (AMV) in the thermal comfort literature. PMV was calculated using the measured indoor environment and occupant-level parameters for the 2179 responses. TSV and PMV were respectively correlated against the Operative Temperature (T_{op}) as abscissa. For an improved visual representation, the abscissa (T_{op}) was binned in increments of 0.5 °C, and the corresponding values of TSV and PMV were averaged for each T_{op} bin; the comparison between TSV and PMV is illustrated in Fig. 12.

The regression was performed using the Weighted Least Squares (WLS) method, which utilizes the respective bin sizes (number of responses) in each T_{op} bin as regression weights; this is done to minimize the impact of low significance datapoints. As observed from the PMV and TSV regression equations in Fig. 12, the slope of the TSV curve

(0.085) is lower than that of the PMV curve (0.199). In other words, the PMV model predicts a unit change in thermal sensation in response to a 5 °C change in T_{op} , while the observed TSV values imply a unit change in thermal sensation in response to a 11.8 °C change in T_{op} . Therefore, the PMV model underpredicts the thermal adaptation of the Indian residential population and should be substituted with an improved, localised model in India Model for Adaptive Comfort – Residential (IMAC-R), as proposed in this study.

4.3.2. Deriving thermal neutralities and acceptability bands

In the process of devising the proposed thermal comfort model, the 2179 responses were divided into 82 bins for ‘Location + Month + Mode’ for the ease of statistical operations, as described in Section 3.3. For instance, as per this binning approach, Bin 1 entails 32 responses from the NV residences in Ahmedabad during January. The statistical summary of the 82 ‘Location + Month + Mode’ bins is provided in Annexure 7.8.

As per the steps described in Section 3.3, ΔTSV and ΔT_{op} were calculated for each response using bin-specific averages using Equation (1) and Equation (2). Fig. 13 shows the unbinned Ordinary Least Squares (OLS) Regression between ΔTSV and ΔT_{op} for the 2179 responses. The slope of the resultant regression curve ($m = 0.151$) was used to calculate the Griffith’s Constant (GC = 0.152) using Equation (3) and Equation (4).

The Calculated GC (0.152) was used to derive bin-specific thermal neutralities using Equation (5). Simultaneously, the bin-specific thermal neutralities were also derived using the GC (0.16) referred from Manu et al. [24]. As stated in Section 3.3, it is essential to note that Manu et al. calculated GC as 0.16 and 0.15 for MM and NV commercial buildings, respectively. However, the study prescribed a common GC (0.16) for devising the adaptive models for MM and NV commercial buildings. A common GC was used, considering its physical significance, the nature of observed data, and the feasibility of model implementation. Consistent with Manu et al., the thermal neutralities for the proposed thermal comfort model were also calculated using GC = 0.16.

Fig. 14 compares the thermal neutralities (T_{neut}) calculated using GC = 0.16 (Manu et al.) and GC = 0.152 (Calculated). The mean difference between the two sets of T_{neut} values remained within 0.1 ± 0.2 °C; this is negligible considering the parameter’s application in thermally dynamic indoor environments. The T_{neut} (for GC = 0.16) was observed to vary between 18.2 and 38.3 °C in response to the T_{op} range of 12.5–33.5 °C. This T_{neut} was combined with 80% and 90% thermal acceptability bands to formulate the final thermal comfort model.

The 80% and 90% thermal acceptability bands were calculated using the sensitivity-slope method, as described in Section 3.3, for the (11 out

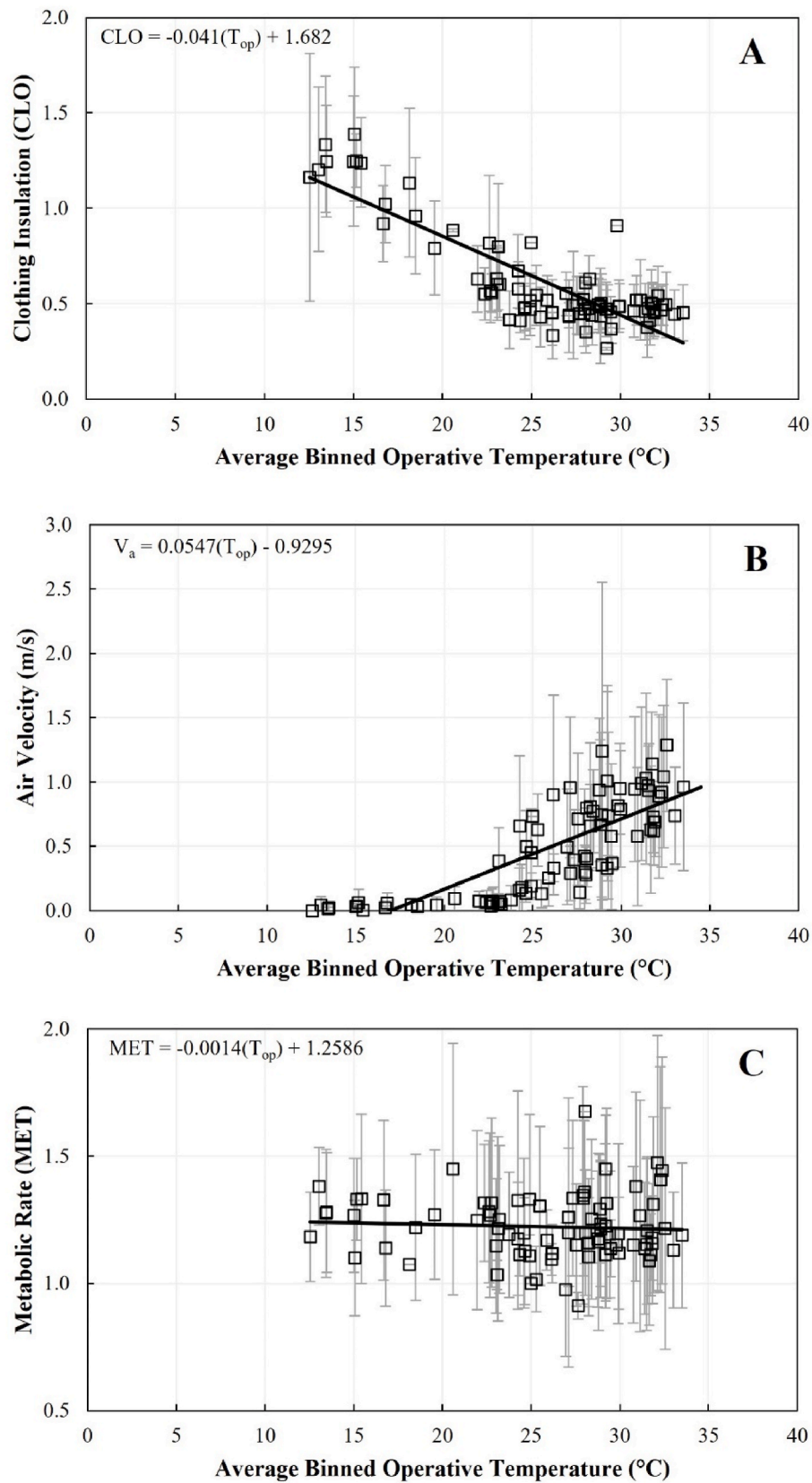


Fig. 8. Adaptive response indicated by the variation of (a) Clothing Insulation (CLO), (b) Indoor Air Velocity (V_a), and (c) Metabolic Rate (MET) with respect to average binned operative temperature.

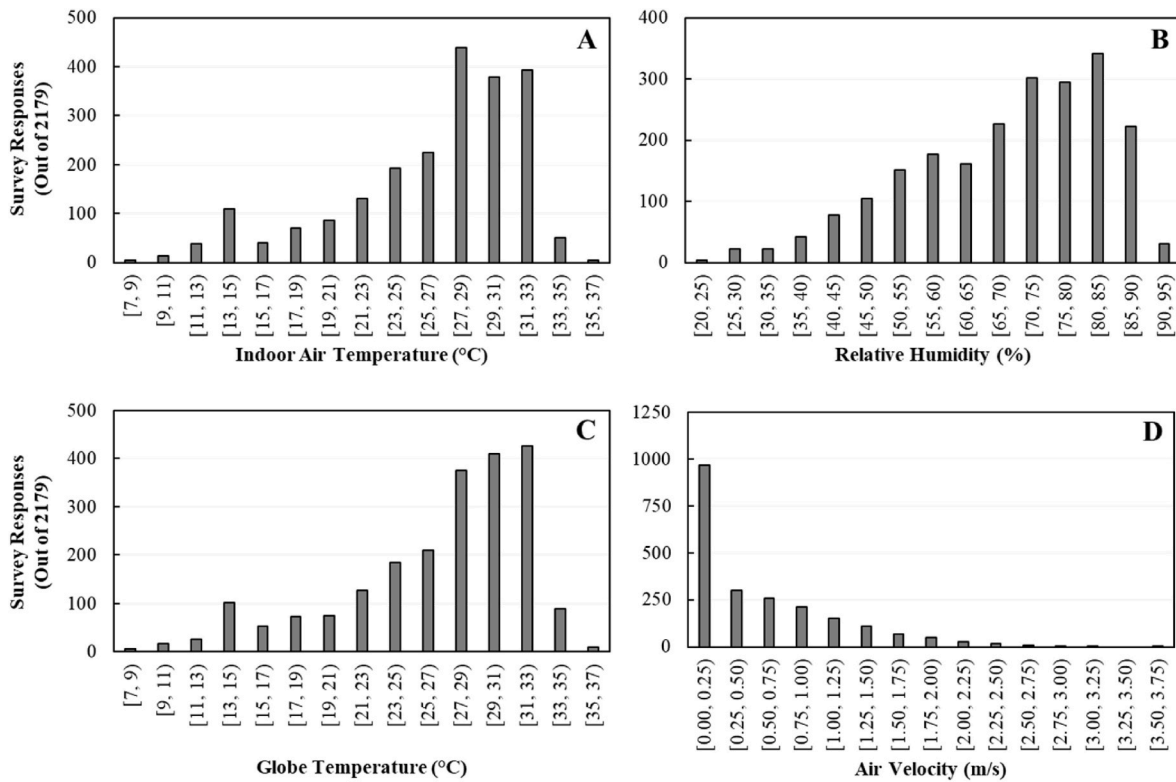


Fig. 9. Distribution of measured indoor (a) Air Temperature (T_{air}), (b) Relative Humidity (RH), (c) Globe Temperature (T_g), and (d) Air Velocity (V_a).

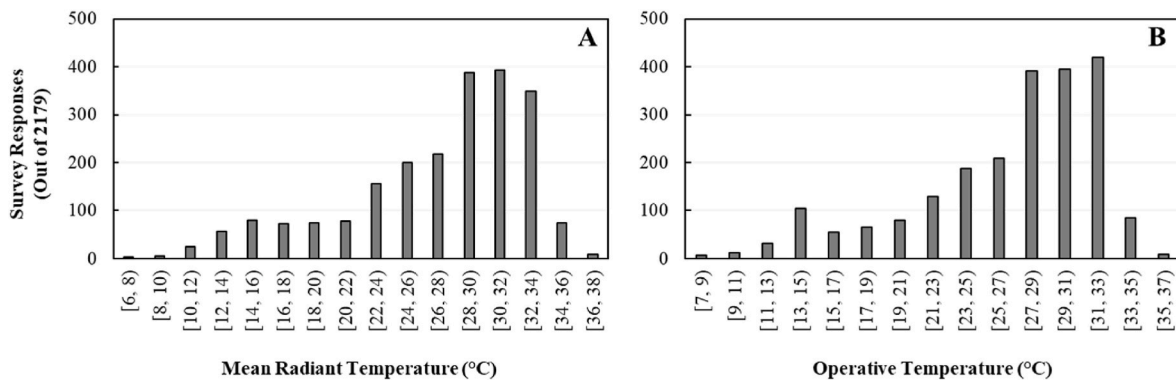


Fig. 10. Calculated indoor (a) mean radiant temperature (MRT) and (b) operative temperature (T_{op}).

of 82) statistically significant 'Location + Month + Mode' bins. The Weighted Average Sensitivity Slope was calculated as 0.23 – this indicates that a typical Indian residential occupant's Thermal Sensation Vote (TSV) changes by 0.23 votes with a unit change in the indoor operative temperature (T_{op}). Annexure 7.10 tabulates the data from the 11 significant bins and their corresponding parameters for calculating the Weighted Average Sensitivity Slope.

Accordingly, the 80% and 90% bands of thermal acceptability were calculated as ± 3.60 °C and ± 2.15 °C, respectively, and combined with the T_{neut} derived using Equation (5) to yield the acceptability bands for the proposed thermal comfort model. T_{neut} was correlated against two $T_{out-rep}$ variables in $T_{out-7DRM}$ and $T_{out-30DRM}$ and the outdoor heat stress variable in UTCI; the final thermal comfort models are presented in the following sub-section.

4.4. Adaptive models for Indian residences

The adaptive thermal comfort mode for Indian residences is

presented in Fig. 15. The two models prescribe 80% and 90% acceptability bands for indoor operative temperature (T_{op}) in correlation with (A) 7-day outdoor running mean temperature ($T_{out-7DRM}$) between 5.5 and 33 °C, and (B) 30-day outdoor running mean temperature ($T_{out-30DRM}$) between 5.5 and 33 °C. These models are applicable to residences with NV or MM modes of building operation. Models A and B demonstrate the relationship between the outdoor representative temperature ($T_{out-7DRM}$ and $T_{out-30DRM}$) and indoor operative temperature (T_{op}).

Additionally, the analysis also explores the variation of T_{neut} with the outdoor heat stress parameter - Universal Thermal Comfort Index (UTCI), in the range of 2–55 °C, as shown in Fig. 16. While dynamic averages of T_{out} (absicca in Fig. 15) attempted to capture the human thermal response to the outdoor air temperature alone, UTCI inherently accounts for six climatic parameters in T_{out} , RH_{out} , outdoor dew-point temperature, solar radiation, outdoor air velocity, and atmospheric pressure that directly and dynamically influence human thermal inclination.

Therefore, a UTCI-based analysis can holistically represent the

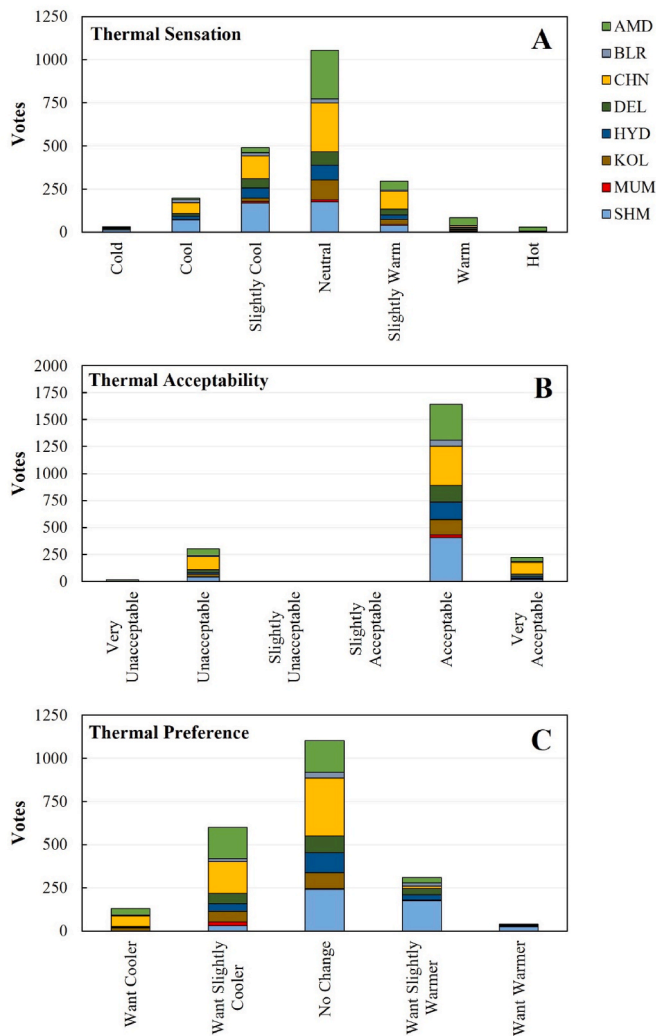


Fig. 11. Observed thermal sensation, acceptability, and preference votes.

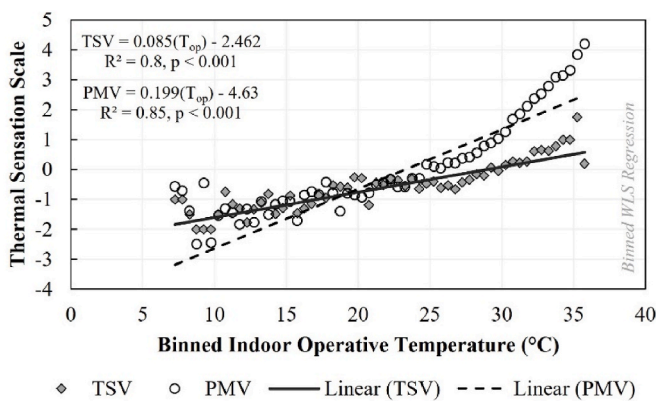


Fig. 12. Comparing observed TSV and calculated PMV with respect to 0.5 °C T_{op} bins.

occupants' dynamic thermal response to the outdoors. However, the relatively complex calculation methodology for UTCI (relying on multiple dependent variables) can be a barrier to its widespread use. Furthermore, none of the existing thermal comfort models follow a UTCI-based analysis, disallowing relevant comparisons. On the other hand, the models illustrated in Fig. 15 can be feasibly compared with the

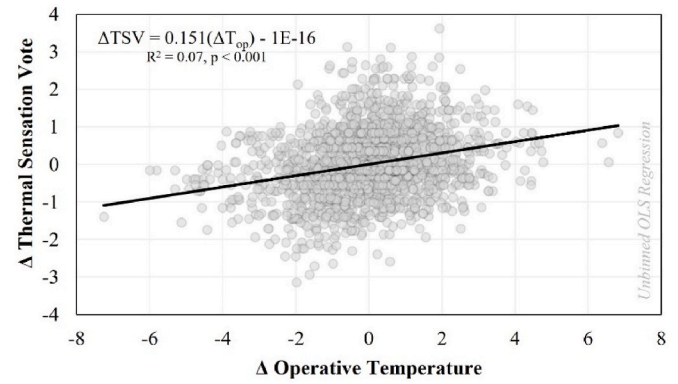


Fig. 13. Δ TSV vs. ΔT_{op} regression for 2179 responses.

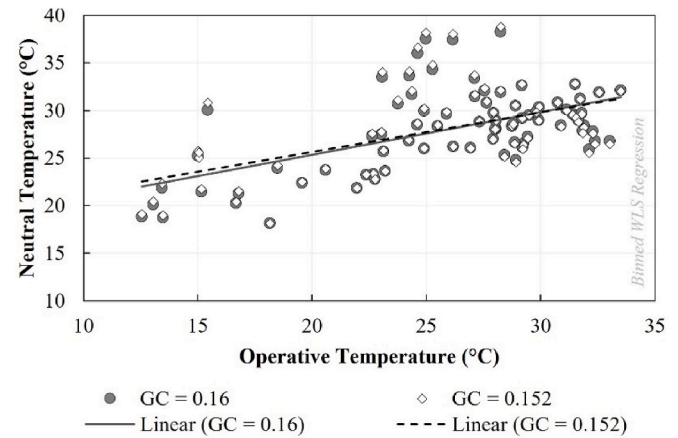


Fig. 14. Thermal neutralities using GC = 0.16 (from IMAC) and GC = 0.152 (calculated) for 82 bins.

proposed model, as demonstrated in the subsequent sub-section.

5. Discussion

The present study proposes the India Model for Adaptive Comfort – Residential (IMAC-R) based on yearlong surveys in eight cities across five Indian climate zones. In this section, the model is compared against the relevant national and international thermal comfort models, and the potential of further research is briefly explored.

5.1. Comparing thermal neutralities

The thermal neutralities of the various national and international models were compared against the neutralities established by the proposed thermal comfort model; in this process, the abscissa of the proposed model was modified as per exponentially weighted values of $T_{out-7DRM}$ and $T_{out-30DRM}$ (with $\alpha = 0.8$, representing the weighting factor [53]) for comparison purposes. Fig. 17 compares the thermal neutrality curve of the proposed model against those of the relevant adaptive models from the literature. Fig. 17(A) compares the proposed model with IMAC (MM), IMAC (NV), and ASHRAE-55, 2020, with the abscissa as $T_{out-30DRM}$; Fig. 17(B) compares the proposed model with EN 16798-1, with the abscissa as $T_{out-7DRM-W}$.

Comparing IMAC-R with the IMAC (MM and NV) models in Fig. 15 (A), it was observed that the occupants in commercial buildings experienced thermal neutrality at relatively lower operative temperatures than the residential occupants. The applicability range of the proposed thermal comfort model (5.5–33 °C) also differs from IMAC (NV: 12.5–31 °C and MM: 13–38.5 °C). Overall, this comparison reflects a

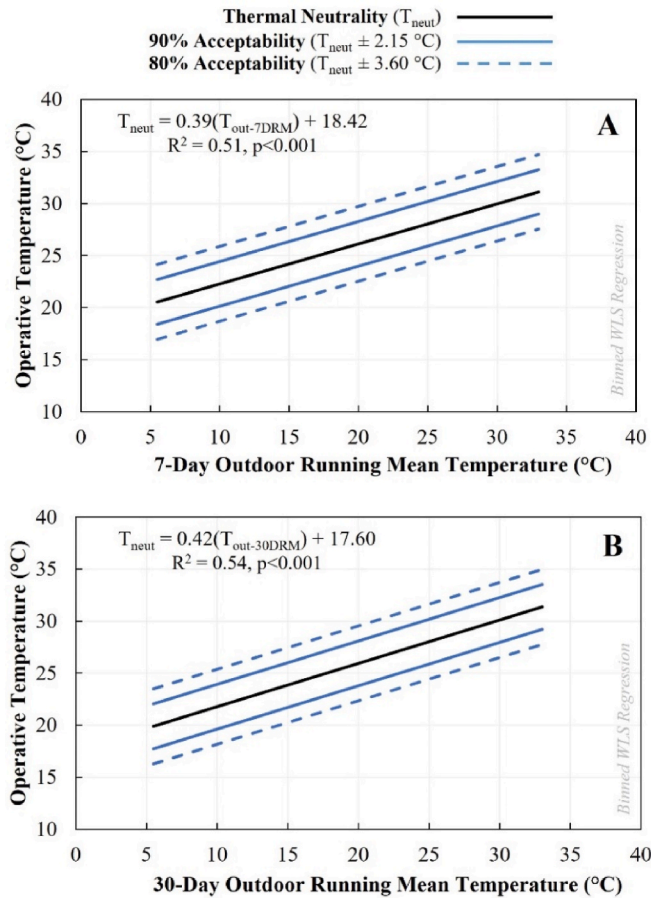


Fig. 15. The proposed adaptive thermal comfort model for Indian residences, with abscissa of (a) $T_{out-7DRM}$ and (b) $T_{out-30DRM}$.

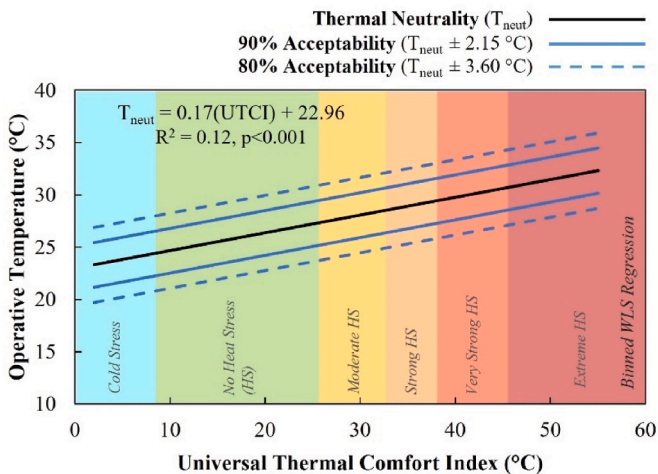


Fig. 16. India Model for Adaptive Comfort – Residential (IMAC-R), with abscissa of UTCI.

more prominent occupant thermal adaptation potential in Indian residences compared to commercial spaces. Further, comparing the model with international models (ASHRAE-55 and EN 16798-1) in Fig. 17(A and B) also reveals that the Indian residential occupants accept warmer indoor environments relative to the global estimates.

IMAC-R was also contrasted against the findings of several regional studies on thermal comfort, as highlighted in Section 2.2 [20–23]. The

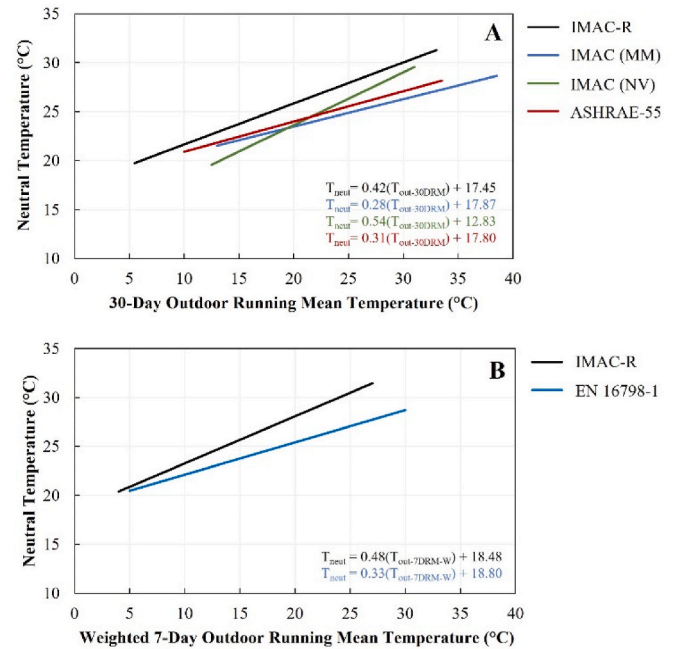


Fig. 17. Comparing the IMAC-R thermal neutralities with IMAC (MM/NV), ASHRAE-55, and EN 16798-1. Note: In Figure B, the IMAC-R neutralities have been rescaled as per the abscissa for comparison.

studies on warmer Indian climates were consistent with IMAC-R in their prescription of the T_{neut} with respect to the $T_{out-rep}$ [21,23]. However, the studies on the cold climate significantly differed in their conclusions at lower values of $T_{out-rep}$. For instance, the regional studies prescribed T_{neut} as $\sim 16^\circ\text{C}$, corresponding to $T_{out-rep}$ of 10°C [20,22,23]. In comparison, the values prescribed by IMAC-R ($T_{neut} = 21.6^\circ\text{C}$), ASHRAE-55 ($T_{neut} = 20.9^\circ\text{C}$), and EN 16798-1 ($T_{neut} = 22.1^\circ\text{C}$) for the same $T_{out-rep}$ were found to be mutually consistent. The comparison between the neutralities of the proposed thermal comfort model and those from the regional studies has been illustrated in Fig. 18.

5.2. Implications and future research

The insights offered by the proposed thermal comfort model complement the existing, nationally recommended model for Indian commercial buildings – India Model for Adaptive Comfort for Commercial Buildings (IMAC), which was incorporated in the Indian National Building Code and its data included as a part of the ASHRAE Global Thermal Comfort Database II [24,25,41]. Through this study, a statistically validated understanding of occupants' comfort trends can serve as crucial evidence for academics, HVAC engineers, building service professionals, architects, national policymakers, and international organizations working on the broad themes of building design, energy efficiency, thermal comfort, and access to cooling.

Future studies can utilize the proposed data collection and analysis methodology to generate standardized regional thermal comfort databases and help develop regional thermal comfort models. A consistent methodology will allow time-agnostic collation of thermal comfort data points for a more comprehensive understanding of national thermal comfort with appropriate regional nuance. Future studies can further explore the applicability potential of envelope-level and occupant-level adaptive strategies, emphasizing the regional climate.

6. Conclusion

This study proposes the India Model for Adaptive Comfort – Residential (IMAC-R) based on a year-long survey of 294 occupied

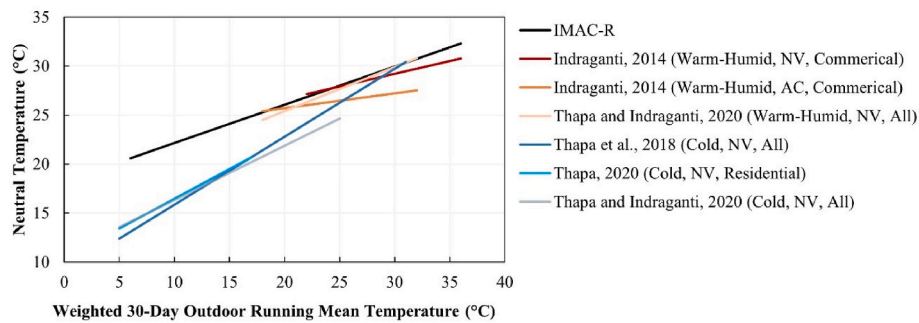


Fig. 18. Comparing the IMAC-R thermal neutralities with the neutralities of the regional models.

residences from five Indian climate zones. The 2179 responses majorly represented the low-rise residences under 25 years of construction age and occupants from diverse age groups (19–62 years). These residences were operated using Mixed-Mode Ventilation or Natural Ventilation. The measurement of indoor environment data and collection of subjective psychological responses helped model the comfortable temperature limits (with 80% and 90% acceptability) for the Indian residential occupants. The comfortable temperature is correlated against the outdoor representative temperatures ($T_{out-7DRM}$ and $T_{out-30DRM}$) and heat stress index (UTCI). The model is largely consistent with the findings of its national and international counterparts. A judicious implementation of this model can help improve energy efficiency and thermal comfort in Indian residences.

CRedit authorship contribution statement

Rajan Rawal: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Yash Shukla:** Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Resources,

Supervision, Writing – original draft, Writing – review & editing. **Vishnu Vardhan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Sneha Asrani:** Data curation, Formal analysis, Investigation, Project administration, Resources, Visualization, Writing – original draft. **Marcel Schweiker:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis. **Richard de Dear:** Conceptualization, Formal analysis, Methodology, Supervision. **Vishal Garg:** Validation, Supervision, Investigation. **Jyotirmay Mathur:** Investigation, Supervision, Validation. **Sanjay Prakash:** Validation, Supervision. **Saurabh Diddi:** Supervision, Validation. **S. Vikash Ranjan:** Validation. **Abdullah N. Siddiqui:** Validation. **Govinda Somani:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

7 Annexure

7.1 Translation of Psychological Voting Scales

Sensation Vote (English)	Hindi	Gujarati	Marathi	Kannada	Tamil	Telugu	Bengali
Cold	बहुत ठंडा	ખૂબજ ઠંડુ	खूप थंड	ತುಂಬಾ ಚಳಿ	கடும்பையான காளிர்	చాలా చలి	খুব ঠান্ডা
Cool	ঠাণ্ডা	ઠંડુ	थंड	ಚಳಿ	காளிர்	చల్లని	শীতল
Slightly cool	थोड़ा ठंडा	સહેજ ઠંડુ	कचिती थंड	ಸೌವಲ್ಪ ಚಳಿ	லசோன காளிர்	కొద్దిగా చలి	কিছুটা ঠাণ্ডা
Neutral	बराबर	બરાબર	बरोबर	ಸಾಮಾನ್ಯ	நடாநிலை	వేడి లేదా చల్లని ఈదు	নিরপেক্ষ
Slightly warm	थोड़ा गर्मी	સહેજ ગરમી	कचिती गरम	ಸೌವಲ್ಪ ಬಿಚ್ಚುಗೆ	லசோன சூலு	కొద్దిగా వేడి	কিছুটা গরম
Warm	गर्मी	ગરમી	गरम	ಬಿಚ್ಚುಗೆ	சூலு	వేడి	উষ্ণ
Hot	बहुत गर्मी	ખૂબજ ગરમી	खूप गरम	ತುಂಬಾ ಶೀತ	கடும்பையான சூலு	చాల వేడిగా	খুব গরম

7.2 Brief Description of Survey Forms

1. Pre-site Checklist

This form was populated by the researcher before each site visit. It was a simple way of ensuring the appropriate operation of the measurement instrumentation and avoiding any functional issues in data collection.

2. Background Form

This form was populated once for a unique residence at the time of the first RNRH survey. It includes the details of the building's typology, material, age, no. of floors, no. of bedrooms, and the occupants' name, gender, age, and years spent in the house. Each occupant and residence were assigned a unique occupant ID and building ID respectively.

3. RNRH Survey Form

This form recorded the occupants' psychological responses to the indoor environment. It collected the parameters of occupants' thermal sensation, thermal preference, thermal acceptance, air movement acceptance, air movement preference, and overall comfort vote, clothing insulation, and

metabolic rate. Along with this, the environmental parameters like Air Temperature (T_a), Relative Humidity (RH), Air Velocity (V_a), and Globe Temperature (T_g) were measured.

7.3

Surveyed Parameters (Background and Recurring)

Parameter	Unit	Survey Form	Frequency	Description
Age Group	Category	Background	Background	Age-group of the occupant
Sex	Category	Background	Background	Gender of the occupant: Male/Female
Acclimatisation Years	Years	Background	Background	No. of years the occupant has spent in the building
Survey Language	Category	Background	Background	Language in which the RNRH survey was conducted.
Building Typology	Category	Background	Background	Typology of the site
Building Material	Category	Background	Background	Walling material of the site
Building Age	Years	Background	Background	Age of the building
Total Floors	No.	Background	Background	Total no. of floors in the building
Site Floor	No.	Background	Background	Floor on which the site is located
Bedrooms	No.	Background	Background	No. of bedrooms in the site
Address	NA	RNRH	Monthly	Address of the house where the RNRH survey was taken. It was a 7-character alphanumeric code, wherein, the first three letters indicate the city (e.g., AMD, CHN, etc.), and the numbers (e.g., A01, A02, etc.) indicate the house no.
Survey Date	Date	RNRH	Monthly	Date on which the survey was taken. Format: DD-MM-YYYY
Survey Time	Time	RNRH	Monthly	Time at which the survey was taken. In a 24-h format: HH:MM:SS
Sex	Category	RNRH	Monthly	Gender of the occupant: Male/Female
Age Group	Category	RNRH	Monthly	Age-group of the occupant
Survey Location	Category	RNRH	Monthly	Occupants' location at the time of the survey
Occupant's Past Location	Category	RNRH	Monthly	Occupants' location 30 min prior to the survey
Thermal Sensation	Vote	RNRH	Monthly	Occupants' Thermal Sensation Vote. It was marked on Fanger's seven-point scale
Thermal Acceptance	Vote	RNRH	Monthly	Occupants' Thermal Acceptance Vote
Thermal Preference	Vote	RNRH	Monthly	Occupants' Thermal Preference Vote
Air Movement Acceptance	Vote	RNRH	Monthly	Occupants' Air Movement Acceptance Vote
Air Movement Preference	Vote	RNRH	Monthly	Occupants' Air Movement Preference Vote
Overall Comfort	Vote	RNRH	Monthly	Occupants' Overall Comfort Vote
Clothing Items	Category	RNRH	Monthly	Clothing items worn by the occupant. Any one of the following options could be selected depending on the type of the clothing material: • Light Weight • Medium Weight • Heavy Weight
Clothing Items (Addition/Removal)	Category	RNRH	Monthly	This cell indicated whether the occupant added/removed any clothing item in the past hour
Seating Type	Category	RNRH	Monthly	Type of seating occupied by the respondent. It was selected from the following: • Chair, Upholstered • Chair, Not Upholstered • Stool • Bed • On floor, without rug • On floor, with rug
Act 10	Category	RNRH	Monthly	Occupant's Activity in the last 10 min (0–10 min)
Act 20	Category	RNRH	Monthly	Occupant's activity in the 10 min before that. (11–20 min)
Act 30	Category	RNRH	Monthly	Occupant's activity in the 10 min before that. (21–30 min)
Act 60	Category	RNRH	Monthly	Occupant's activity in the 30 min before that. (31–60 min)
Air Temperature	°C	RNRH	Monthly	Air Temperature of the space where the RNRH survey was taken
Relative Humidity	%	RNRH	Monthly	Relative Humidity of the space where the RNRH survey was taken
Air Velocity in X Axis	m/s	RNRH	Monthly	Air Velocity in X axis of the space where the RNRH survey was taken
Air Velocity in Y Axis	m/s	RNRH	Monthly	Air Velocity in Y axis of the space where the RNRH survey was taken
Air Velocity in Z Axis	m/s	RNRH	Monthly	Air Velocity in Z axis of the space where the RNRH survey was taken
Globe Temperature	°C	RNRH	Monthly	Globe Temperature of the space where the RNRH survey was taken
Air Conditioner Presence	Binary	RNRH	Monthly	Indicates the presence of an AC in the surveyed space
Air Conditioner Control	Binary	RNRH	Monthly	Indicates whether the occupant had control over the AC or not
Air Conditioner Status	Binary	RNRH	Monthly	Indicates whether the AC was switched on at the time of the survey or not
Occupant's Distance from Air Conditioner	Meter	RNRH	Monthly	Distance between the AC and the occupant
Air Cooler Presence	Binary	RNRH	Monthly	Indicates the presence of Air cooler in the surveyed space
Air Cooler Control	Binary	RNRH	Monthly	Indicates whether the occupant had control over the Air Cooler or not
Air Cooler Status	Binary	RNRH	Monthly	Indicates whether the Air Cooler was switched on at the time of the survey or not
Occupant's Distance from Air Cooler	Metre	RNRH	Monthly	Distance between the Air Cooler and the occupant.
Fan Presence	Binary	RNRH	Monthly	Indicates the presence of a fan in the surveyed space
Fan Type	Binary	RNRH	Monthly	Indicates the type of fan. Yes = ceiling fan No = not a ceiling fan, e.g., table fan, wall mounted fan etc.
Fan Control	Binary	RNRH	Monthly	Indicates whether the occupant had control over the fan or not.
Fan Status	Binary	RNRH	Monthly	Indicates whether the fan was switched on at the time of the survey or not
Occupant's Distance from Fan	Meter	RNRH	Monthly	Distance between the fan and the occupant
Window Presence	Binary	RNRH	Monthly	Indicates the presence of a window in the surveyed space
Window Control	Binary	RNRH	Monthly	Indicates whether the occupant had control over the window or not
Window Status (Partially)	Binary	RNRH	Monthly	Indicates whether the window was partially open or not

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7.3 (continued)

Parameter	Unit	Survey Form	Frequency	Description
Window Status (Fully)	Binary	RNRH	Monthly	Indicates whether the window was fully open or not
Occupant's Distance from Window	Meter	RNRH	Monthly	Distance between the window and the occupant
Heater Presence	Binary	RNRH	Monthly	Indicates the presence of a heater in the surveyed space
Heater Control	Binary	RNRH	Monthly	Indicates whether the occupant had control over the heater or not
Heater Status	Binary	RNRH	Monthly	Indicates whether the heater was switched on at the time of the survey or not
Occupant's Distance from Heater	Meter	RNRH	Monthly	Distance between the heater and the occupant

7.4

Integrated Surface Database (ISD) Description

Parameter	Unit	Description
Year	Year	Year of the data
Month	Month	Month of the data
Day	Day	Day of the data
Hour	Hour	Hour of the data
Minute	Minute	Minute of the data
Dry Bulb Temperature	°C	Dry Bulb Temperature at the time indicated. Range: 70 to 70
Dew Point Temperature	°C	Dew Point Temperature at the time indicated. Range: 70 to 71
Relative Humidity	%	Relative Humidity at the time indicated. Range: 0 to 110
Atmospheric Station Pressure	Pa	Station Pressure at the time indicated. Range: 31,000 to 120,000
Extra-terrestrial Horizontal Radiation	Wh/m ²	Extra-terrestrial Horizontal Radiation at the time indicated. Minimum value: 0
Extra-terrestrial Direct Normal Radiation	Wh/m ²	Extra-terrestrial Direct Normal Radiation is the amount of solar radiation received on a surface normal to the rays of the sun at the top of the atmosphere during the number of minutes preceding the time indicated
Horizontal Infrared Radiation Intensity	Wh/m ²	Horizontal Infrared Radiation intensity at the time indicated. Minimum value: 0
Global Horizontal Radiation	Wh/m ²	Total amount of Direct and Diffuse Solar Radiation received on a horizontal surface during the number of minutes preceding the time indicated
Direct Normal Radiation	Wh/m ²	Amount of Solar Radiation received directly from the solar disk on a surface perpendicular to the sun's rays, during the number of minutes preceding the time indicated
Diffuse Normal Radiation	Wh/m ²	Amount of Solar Radiation received from the sky (excluding the solar disk) on a horizontal surface during the number of minutes preceding the time indicated
Global Horizontal Illuminance	lux	Average total amount of Direct and Diffuse Illuminance received on a horizontal surface during the number of minutes preceding the time indicated
Direct Normal Illuminance	lux	Average amount of Illuminance received directly from the solar disk on a surface perpendicular to the sun's rays, during the number of minutes preceding the time indicated.
Diffuse Horizontal Illuminance	lux	Average amount of Illuminance received from the sky (excluding the solar disk) on a horizontal surface during the number of minutes preceding the time indicated
Zenith Luminance	Cd/m ²	Average amount of Luminance at the sky's zenith during the number of minutes preceding the time indicated
Wind Direction	Degrees	Wind Direction observed at the time indicated, where the convention is that North = 0.0, East = 90.0, South = 180.0, West = 270.0
Wind Speed	m/s	Wind Speed observed at the indicated time
Total Sky Cover	tenths	Amount of sky dome covered by clouds or obscuring phenomena that prevent observing the sky or higher cloud layers at the time indicated
Opaque Sky Cover	tenths	Amount of sky dome covered by clouds or obscuring phenomena that prevent observing the sky or higher cloud layers at the time indicated
Visibility	km	Visibility observed at the indicated time
Ceiling Height	m	Ceiling height observed at the indicated time
Present Weather Observation	Category	Present weather observed, primarily concerning rain/wet surfaces, at indicated time
Present Weather Codes	Category	Present weather codes, primarily concerning rain/wet surfaces, at indicated time
Precipitable Water	mm	Precipitable water observed at the indicated time
Aerosol Optical Depth	thousandths	Aerosol optical depth observed at the indicated time
Snow Depth	cm	Snow depth observed at the indicated time
Days Since Last snowfall	Day	Days since last snowfall observed at the indicated time
Albedo	unitless	Ratio of reflected solar irradiance to global horizontal irradiance
Liquid Precipitation Depth	mm	Amount of liquid precipitation observed at the indicated time for the period indicated in the liquid precipitation quantity field
Liquid Precipitation Quantity	hour	Accumulation period for the liquid precipitation depth field

7.5

City-specific Measured Parameters and Occupant Responses

City	Observed Range of the Indoor Parameters			
	Air Temperature (°C)	Relative Humidity (%)	Air Velocity (m/s)	Globe Temperature (°C)
AMD	AC ON: 20–30 °C AC OFF: Daily T _{out} Range ± 3 °C	AC ON/OFF: Daily RH _{out} Range ± 18%	AC ON: 0–3 m/s FAN OFF: 0–0.5 m/s FAN ON: 0–2.5 m/s	AC ON: T _a - T _g < 2.5 °C AC OFF: T _a - T _g < 4.5 °C
BLR	AC ON/OFF: Daily T _{out} Range	AC ON/OFF: Daily RH _{out} Range	FAN OFF: 0–0.5 m/s	AC ON/OFF: T _a - T _g < 1.5 °C

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7.5 (continued)

City	Observed Range of the Indoor Parameters								
	Air Temperature (°C)	Relative Humidity (%)	Air Velocity (m/s)	Globe Temperature (°C)					
CHN	AC ON: 20–30 °C AC OFF: Daily T _{out} Range ± 3 °C	AC ON: Daily RH _{out} Range ± 16% AC OFF: Daily RH _{out} Range ± 6%	FAN ON: 0–1 m/s AC ON: 0–2 m/s FAN OFF: 0–0.5 m/s FAN ON: 0–3 m/s	AC ON: T _a - T _g < 3.5 °C AC OFF: T _a - T _g < 1.5 °C					
DEL	AC ON: 23–35 °C AC OFF: Daily T _{out} Range ± 2.5 °C	AC ON: Daily RH _{out} Range ± 30% AC OFF: Daily RH _{out} Range ± 10%	AC ON: 0–1 m/s FAN OFF: 0–0.5 m/s FAN ON: 0–2 m/s	AC ON: T _a - T _g < 2.5 °C AC OFF: T _a - T _g < 4.5 °C					
HYD	AC ON: 24–28 °C AC OFF: Daily T _{out} Range ± 1.5 °C	AC ON/OFF: Daily RH _{out} Range ± 15%	AC ON: 0–0.5 m/s FAN OFF: 0–1.2 m/s FAN ON: 0–2 m/s	AC ON/OFF: T _a - T _g < 2.5 °C					
KOL	AC ON: 22–30 °C AC OFF: Daily T _{out} Range ± 3.5 °C	AC ON: Daily RH _{out} Range ± 15% AC OFF: Daily RH _{out} Range ± 12%	AC ON: 0–2.5 m/s FAN OFF: 0–1 m/s FAN ON: 0–2.5 m/s	AC ON/OFF: T _a - T _g < 3.5 °C					
MUM	AC ON/OFF: Daily T _{out} Range	AC ON/OFF: Daily RH _{out} Range ± 10%	FAN ON/OFF: 0–2 m/s	AC ON/OFF: T _a - T _g < 1 °C					
SHM	AC ON/OFF: Daily T _{out} Range ± 10 °C	AC ON/OFF: Daily RH _{out} Range ± 10%	FAN OFF: 0–0.5 m/s	AC ON/OFF: T _a - T _g < 4 °C					
Thermal Sensation Scale		Percentage of Total Votes by City							
		AMD	BLR	CHN	DEL	HYD	KOL	MUM	SHM
Cold		0%	0%	0%	0%	0%	0%	0%	1%
Cool		0%	1%	3%	1%	1%	0%	0%	3%
Slightly Cool		1%	1%	6%	2%	3%	1%	0%	8%
Neutral		13%	1%	13%	3%	4%	5%	1%	8%
Slightly Warm		2%	0%	5%	2%	1%	1%	0%	2%
Warm		2%	0%	0%	0%	0%	0%	0%	0%
Hot		1%	0%	0%	0%	0%	0%	0%	0%
Thermal Acceptability Scale		Percentage of Total Votes by City							
		AMD	BLR	CHN	DEL	HYD	KOL	MUM	SHM
Very Unacceptable		1%	0%	0%	0%	0%	0%	0%	0%
Unacceptable		3%	1%	5%	1%	1%	1%	0%	2%
Slightly Unacceptable		0%	0%	0%	0%	0%	0%	0%	0%
Slightly Acceptable		0%	0%	0%	0%	0%	0%	0%	0%
Acceptable		15%	3%	17%	7%	7%	7%	1%	19%
Very Acceptable		2%	0%	5%	1%	1%	0%	0%	1%
Thermal Preference Scale		Percentage of Total Votes by City							
		AMD	BLR	CHN	DEL	HYD	KOL	MUM	SHM
Want Cooler		2%	0%	3%	0%	0%	1%	0%	0%
Want Slightly Cooler		8%	1%	8%	3%	2%	3%	1%	1%
No Change		8%	1%	15%	4%	5%	4%	0%	11%
Want Slightly Warmer		1%	1%	1%	2%	1%	0%	0%	8%
Want Warmer		0%	0%	0%	0%	0%	0%	0%	1%

7.6 Data Cleaning Details

1. Unique Response ID

Each response was given a unique response ID containing the details of the city, house, occupant ID and date of the survey (e.g., SHM_46_13_10_2020_1 m).

2. Flagging Duplicate Responses

- By applying conditional formatting to the column of response ID, the responses with identical response IDs were flagged.
- The survey date and time of the duplicate responses were checked to understand if it was a case of either of the following:
 - a. Submission Error – A single RNRH form being submitted twice
 - b. Data Entry Error – A manual error of entering the wrong occupant ID (e.g., entering 1 M instead of 1F)
- The data entry error (in occupant ID) was rectified manually, considering the following factors:
 - a. Deciding factor for gender – clothing and activity
 - b. Comparing the age group and gender details from the background data
 - c. Inquiring with the surveying researcher about the error

3. Date Format

- The survey date column was filtered manually to check the dates/months of data collection.
- Since the survey was filled on Google forms, the date format varied, e.g., for survey taken on 8 March 2021, the date would be entered as 03/08/21 (mm/dd/yyyy) but interpreted as 3 August 2021. Knowing that there was no data collection in August, this data entry was flagged, and its format checked.
- The survey date for preceding surveys was checked manually to rectify this error.

4. Time Format

- In the form, the survey time was entered in 12-h format, and AM/PM was selected from a dropdown.
- This check was done manually by filtering the 'survey time' column.
- The surveys were conducted between 9 a.m. and 9 p.m., any data entry outside this boundary was identified and rectified manually (e.g., the survey date entry was 4:30 a.m., logically it was understood that the survey was taken at 4:30 p.m.)

5. Occupant ID Validation

- Error in OCC_ID Nomenclature:
 - a. Simple errors like changing "F1" to "1F", were identified, flagged, and changed manually, maintaining the Occ_ID nomenclature.
- Error in OCC_ID:
 - a. Error in age group (numeric part of nomenclature) at places, the Occ_ID was wrongly entered, for e.g., as 1F, whereas the response was taken from 2F.
 - b. Error in gender (alphabetic part of nomenclature) for e.g., data entry was 1F, but the survey was taken by 1 M.
 - c. Such occurrences were flagged manually based on identical response IDs or based on a difference in the gender data entry and the alphabetic part of the OCC_ID (M/F).
 - d. This error was also rectified by checking the gender and age-group.

6. Check for Gender

- The gender of the respondent as indicated by the alphabetical character of Occ_ID, i.e., M for male, F for female, and that selected in the survey form, did not match.
- To flag this error, an "if" formula was used to compare the last letter of the occupant ID (M/F) and first letter of the gender (M/F) data entry.
- This error was manually corrected based on previously submitted responses of the same site of the same day, clothing, and activity of the respondent.

7. Check for Age Group

This check was done to ensure that the respondent's age group entered in the RNRH survey and background survey.

8. Check for Blank Cells

This check was done automatically using an excel if formula. The data entries for the following column headers were checked for null values, because a null value would lead to that whole dataset (1 row) being discarded:

- Thermal Sensation Vote
- Thermal Acceptance Vote
- Thermal Preference Vote
- Air Movement Acceptance Vote
- Air Movement Preference Vote
- Clothing Items
- Metabolic Activity
- Indoor Environmental Parameters

9. Check for Clothing

- To ensure that appropriate clothing items were marked, this check was done automatically using Excel formula. This formula listed all the clothing items marked in the form.
- This list was then checked manually by filtering the column, to look for the following errors:
 - a. Where all the clothing items were ticked/checked
 - b. Banyan/under-shirt being marked for the female
 - c. While marking sari, petticoat and/or blouse being unmarked; these missing entries were added
 - d. The marked clothing items constituted of an incomplete clothing assembly (the lower or the upper clothing missing), e.g., Stockings, T-shirt, Shoes, and Socks.
- This error was rectified by logically removing/adding relevant clothing items, e.g., adding 'light weight-pants' in case if only t-shirt/shirt was marked.

10. Check for Metabolic Activity

- The metabolic activity was marked for four-time brackets. At places, for a single time bracket, more than one metabolic activity was selected.
- The error was flagged automatically using Excel lookup formula. A code was assigned to each activity, the lookup formula would print this code if only one activity was selected. The formula would print "error" in case of more than one activity.
- This error was dealt with in the following ways:
 - a. Based on the activity marked in the rest of the time brackets, the 'extra' faulty entry was removed, e.g., "seated, working, intense physical activity" were entered in one time bracket, while the rest were marked as "seated, working", then the "intensive physical activity" was removed.
 - b. As the survey form was filled up on a cell phone, and the activity list needed to be scrolled horizontally (image on the left), many times, the first activity – "seated, quiet", was marked by mistake. While removing illogical data entries, this was also taken into consideration.

11. Check for Air Temperature

- This parameter was validated automatically using the 'if' formula in Excel.
- The measured air temperature was checked to satisfy the decided validation range.
 - a. The outdoor daily min and max air temperature were taken from TMY + HOBO files.
 - b. For each city, the validation range was set by studying the range of data, under categories like AC on, AC off, fan on, fan off, window open, window closed etc.
 - c. Visibly erroneous data entries (like $T_a = 136$) was flagged and corrected using the images of the device screens taken during the survey.

12. Check for Relative Humidity

- This parameter was validated automatically using the 'if' formula in Excel.
- The measured air temperature was checked to satisfy the decided validation range.
 - a. The outdoor daily min and max RH were taken from TMY + HOBO files.
 - b. For each city, the validation range was set by studying the range of data, under categories like AC on, AC off, fan on, fan off, window open, window closed etc.
 - c. Visible erroneous data entries (like $RH = 7$) was flagged and corrected using the images of the device screens taken during the survey.

13. Check for Air Velocity

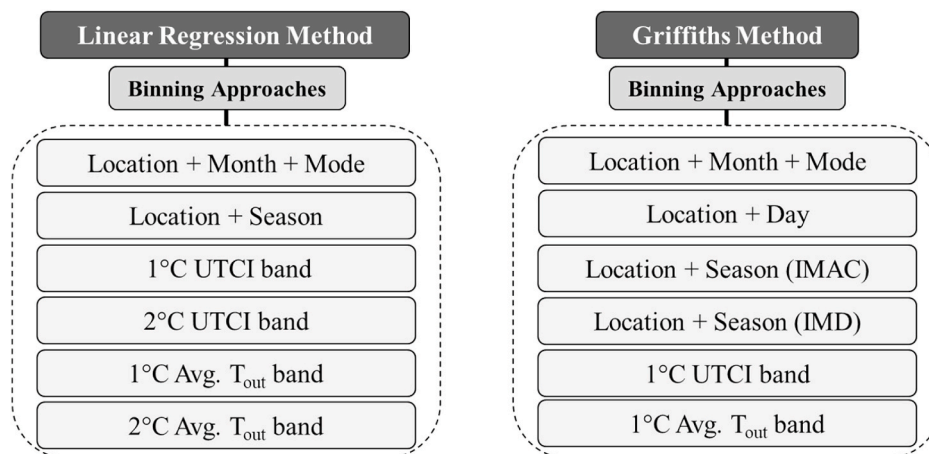
- This error was first checked automatically (using excel formula), then manually. The excel formula ensured that the measured velocity was within the validation range. If not, an "error" message was displayed.
- This "error" was highlighted, and then manually corrected as per the following:
 - a. Any mistake made during data entry in terms of where (T_a or RH were entered instead of velocity) the velocity was entered.
 - b. Entries like "000" were corrected using the images of the device screens taken during the survey.
 - c. Every 10th row was manually checked to confirm that the data entry and the image of the measurement corroborate.

14. Check for Globe Temperature

- Firstly, the image of the measurement was checked to ensure that the correct value was entered. The data entry and measurement image at every 5th row were checked manually to ensure that the Extech HT 200 measured ' T_g ' and not any other parameter (Air, Wet Bulb, or Dewpoint Temperature). All the data entries except T_g were flagged and had to be removed.
- Going by the thumb rule that the difference between air and globe temperature is not very significant (different value for each city/condition), the data entries for T_g were flagged for error. This was done automatically, using an Excel formula. The measurement images for the erroneous entries were rechecked manually. Furthermore, the status of AC, fan, window, cooler and heater were checked to understand if there were a logical/significant explanation for the error.
- Another step included calculating the difference between T_a and MRT. Wherever $T_a - MRT > 5^\circ\text{C}$, those datasets were flagged. Considering the TSV, T_a and MRT of flagged datasets, the erroneous ones were removed.

7.7 Explored Analysis Approaches

Out of the following 12 approaches, the Griffith's Method with 'Location + Month + Mode' Bins was used for model formulation. Furthermore, the same binning approach, combined with Linear Regression Model was used to prescribe 80% and 90% thermal acceptability bands using the sensitivity slope method.



1. Models using Linear Regression Method:

Binning Approach	Bins	Significant Bins	Significant Votes (Total 2179)	Analysis Sub-Method	Final Model	R ²	p < 0.01
Location + Month	57	11	31%	No sub-binning	$T_{neut} = -0.585(T_{30DRM}) + 44.615$	0.18	Yes
		8	22%	0.5°C T _{op} sub-binning	$T_{neut} = 0.623(T_{30DRM}) + 11.539$	0.16	No
Location + Season	21	9	62%	No sub-binning	$T_{neut} = -0.003(T_{30DRM}) + 27.874$	0.37	Yes
		9	62%	0.5°C T _{op} sub-binning	$T_{neut} = -0.01(T_{30DRM}) + 28.082$	0.81	Yes
1°C UTCI band	56	12	46%	No sub-binning	$T_{neut} = -1.104(T_{30DRM}) + 59.209$	0.81	Yes
		11	39%	0.5°C T _{op} sub-binning	$T_{neut} = -1.461(T_{30DRM}) + 68.231$	0.93	Yes
2°C UTCI band	29	8	63%	No sub-binning	$T_{neut} = -1.461(T_{30DRM}) + 68.231$	0.02	No
		7	58%	0.5°C T _{op} sub-binning	$T_{neut} = -1.358(T_{30DRM}) + 65.703$	0.02	Yes
1°C Avg. T _{out} band	27	7	57%	No sub-binning	$T_{neut} = -0.056(T_{30DRM}) + 30.141$	0.02	No
		6	46%	0.5°C T _{op} sub-binning	$T_{neut} = -0.076(T_{30DRM}) + 30.643$	0.01	No
2°C Avg. T _{out} band	14	4	60%	No sub-binning	$T_{neut} = -0.056(T_{30DRM}) + 30.141$	0.18	Yes
		4	60%	0.5°C T _{op} sub-binning	$T_{neut} = -0.045(T_{30DRM}) + 29.842$	0.16	0.49

2. Models using Griffith's Method:

Binning Approach	Bins	Observations	Analysis Sub-Method	Final Model	R ²	p < 0.01
Location + Month + Mode	82	2179 100% Data Acceptable	Calculated GC (0.151)	$T_{neut} = 0.398(T_{30DRM}) + 18.143$	0.49	Yes
			GC = 0.16 (IMAC-C, MM)	$T_{neut} = 0.417(T_{30DRM}) + 17.597$	0.54	Yes
Location + Day	350		Calculated GC (0.145)	$T_{neut} = 0.380(T_{30DRM}) + 18.620$	0.33	Yes
			GC = 0.16 (IMAC-C, MM)	$T_{neut} = 0.418(T_{30DRM}) + 17.590$	0.41	Yes
Location + Season (IMAC-C)	21		Calculated GC (0.153)	$T_{neut} = 0.426(T_{30DRM}) + 17.389$	0.59	Yes
			GC = 0.16 (IMAC-C, MM)	$T_{neut} = 0.410(T_{30DRM}) + 17.833$	0.62	Yes
Location + Season (IMD)	25		Calculated GC (0.165)	$T_{neut} = 0.431(T_{30DRM}) + 17.259$	0.65	Yes
			GC = 0.16 (IMAC-C, MM)	$T_{neut} = 0.443(T_{30DRM}) + 16.938$	0.63	Yes
1°C UTCI band	55		Calculated GC (0.071)	$T_{neut} = 0.345(T_{30DRM}) + 19.358$	0.15	Yes
			GC = 0.16 (IMAC-C, MM)	$T_{neut} = -0.304(T_{30DRM}) + 36.661$	0.58	Yes
1°C Avg. T _{out} band	29		Calculated GC (0.117)	$T_{neut} = 0.414(T_{30DRM}) + 17.686$	0.55	Yes
			GC = 0.16 (IMAC-C, MM)	$T_{neut} = 0.259(T_{30DRM}) + 21.899$	0.84	Yes

7.8

Statistical Parameter Summary for 'Location + Month + Mode' Bins

Bin	Responses	Location	Month	Space Operation Mode	TSV	T _a (°C)	RH (%)	V _a (m/s)	T _{op} (°C)	CLO	MET	Daily Avg. T _{out} (°C)	T _{out-30DRM} (°C)
1	32	AMD	1	NV	-0.41 ± 0.95	24.14 ± 1.16	78.39 ± 2.49	0.16 ± 0.18	24.25 ± 1.05	0.67 ± 0.19	1.33 ± 0.43	19.43 ± 1.03	20.14 ± 0.31
2	68	AMD	2	NV	0.15 ± 0.76	27.65 ± 1.34	81.44 ± 2.79	0.42 ± 0.52	27.92 ± 1.42	0.52 ± 0.14	1.35 ± 0.43	23.24 ± 3.46	20.75 ± 0.79
3	81	AMD	3	NV	0.40 ± 0.85	30.67 ± 1.89	70.44 ± 23.53	0.58 ± 0.54	30.91 ± 1.89	0.52 ± 0.13	1.38 ± 0.37	28.14 ± 1.85	25.46 ± 1.12
4	12	AMD	4	NV	1.00 ± 1.13	31.73 ± 0.83	84.85 ± 1.47	0.89 ± 0.64	32.12 ± 0.91	0.54 ± 0.15	1.47 ± 0.50	32.24 ± 0.00	30.14 ± 0.00
5	39	AMD	9	NV	0.90 ± 1.05	31.93 ± 0.90	84.70 ± 3.46	1.04 ± 0.55	32.37 ± 0.83	0.50 ± 0.11	1.44 ± 0.45	30.71 ± 0.68	29.09 ± 0.91
6	84	AMD	10	NV	0.71 ± 1.20	31.92 ± 1.60	85.27 ± 3.84	0.92 ± 0.58	32.28 ± 1.63	0.47 ± 0.12	1.41 ± 0.44	29.14 ± 1.79	30.52 ± 0.39
7	34	AMD	8	NV	0.44 ± 1.02	28.81 ± 1.25	85.20 ± 4.84	0.74 ± 0.65	29.26 ± 1.27	0.47 ± 0.13	1.31 ± 0.37	27.65 ± 1.22	29.48 ± 0.81
8	25	AMD	7	NV	0.56 ± 1.00	31.51 ± 1.15	75.19 ± 4.61	0.69 ± 0.27	31.90 ± 1.17	0.46 ± 0.13	1.31 ± 0.35	31.01 ± 0.71	30.87 ± 0.10
9	9	AMD	6	NV	0.22 ± 0.67	32.87 ± 1.97	64.37 ± 7.74	0.96 ± 0.65	33.51 ± 1.94	0.45 ± 0.15	1.19 ± 0.29	32.48 ± 1.29	32.13 ± 0.17

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7.8 (continued)

Bin	Responses	Location	Month	Space Operation Mode	TSV	T _a (°C)	RH (%)	V _a (m/ s)	T _{op} (°C)	CLO	MET	Daily Avg. T _{out} (°C)	T _{out-30DRM} (°C)
10	7	AMD	12	NV	−0.71 ± 0.95	22.97 ± 0.96	78.31 ± 2.08	0.06 ± 0.02	23.04 ± 1.01	0.63 ± 0.17	1.15 ± 0.26	15.78 ± 0.00	21.40 ± 0.00
11	6	BLR	11	NV	−1.17 ± 1.47	24.38 ± 0.75	63.90 ± 3.34	0.18 ± 0.18	24.37 ± 0.58	0.41 ± 0.06	1.11 ± 0.08	22.73 ± 0.00	23.59 ± 0.00
12	15	BLR	10	NV	−0.60 ± 1.06	25.76 ± 0.82	72.85 ± 2.29	0.26 ± 0.22	25.88 ± 0.97	0.52 ± 0.13	1.17 ± 0.12	23.63 ± 1.19	23.97 ± 0.07
13	9	BLR	12	NV	−1.11 ± 1.62	23.43 ± 0.60	82.48 ± 0.69	0.08 ± 0.11	23.76 ± 0.45	0.42 ± 0.15	1.19 ± 0.25	21.31 ± 0.13	22.27 ± 0.06
14	25	BLR	1	NV	−0.80 ± 1.19	24.74 ± 1.10	59.29 ± 6.30	0.19 ± 0.10	24.91 ± 0.94	0.52 ± 0.15	1.33 ± 0.33	22.74 ± 0.21	21.76 ± 0.07
15	13	BLR	2	NV	−0.62 ± 1.50	24.43 ± 1.09	49.58 ± 8.93	0.13 ± 0.07	24.61 ± 1.06	0.48 ± 0.12	1.20 ± 0.29	23.54 ± 1.85	22.12 ± 0.26
16	9	BLR	3	NV	0.67 ± 0.87	28.37 ± 0.66	30.42 ± 3.34	0.36 ± 0.27	28.92 ± 0.58	0.50 ± 0.16	1.21 ± 0.27	26.08 ± 0.00	24.93 ± 0.00
17	69	CHN	12	NV	−0.72 ± 0.91	27.26 ± 0.82	77.38 ± 6.97	0.71 ± 0.51	27.55 ± 0.84	0.49 ± 0.15	1.15 ± 0.24	26.48 ± 0.85	26.57 ± 0.20
18	60	CHN	8	NV	0.28 ± 0.76	31.19 ± 0.93	68.32 ± 5.40	1.03 ± 0.67	31.39 ± 0.84	0.46 ± 0.15	1.14 ± 0.26	29.98 ± 0.65	29.68 ± 0.06
19	74	CHN	9	NV	−0.01 ± 0.84	30.48 ± 1.16	74.72 ± 3.97	0.95 ± 0.56	30.75 ± 1.13	0.46 ± 0.14	1.15 ± 0.31	28.63 ± 0.98	29.00 ± 0.15
20	10	CHN	7	NV	−0.20 ± 0.79	31.33 ± 0.58	68.12 ± 2.52	0.97 ± 0.30	31.51 ± 0.82	0.38 ± 0.16	1.15 ± 0.34	30.89 ± 0.00	29.93 ± 0.00
21	45	CHN	2	NV	0.16 ± 0.52	29.55 ± 1.18	62.18 ± 8.03	0.95 ± 0.35	29.92 ± 1.15	0.49 ± 0.14	1.20 ± 0.35	26.25 ± 0.66	25.90 ± 0.13
22	51	CHN	3	NV	0.35 ± 0.52	31.18 ± 1.02	68.94 ± 5.09	0.93 ± 0.36	31.55 ± 1.05	0.48 ± 0.13	1.21 ± 0.29	29.19 ± 1.27	27.47 ± 0.64
23	70	CHN	11	NV	−0.59 ± 0.75	28.03 ± 0.71	81.76 ± 6.83	0.81 ± 0.49	28.25 ± 0.69	0.48 ± 0.13	1.11 ± 0.23	27.34 ± 0.69	27.55 ± 0.39
24	77	CHN	10	NV	−0.55 ± 1.06	28.97 ± 1.56	76.72 ± 4.93	1.01 ± 0.69	29.19 ± 1.68	0.47 ± 0.14	1.23 ± 0.32	27.63 ± 1.39	28.88 ± 0.13
25	59	CHN	1	NV	0.07 ± 0.52	28.42 ± 0.96	73.18 ± 7.42	0.94 ± 0.56	28.76 ± 0.96	0.49 ± 0.14	1.17 ± 0.27	26.22 ± 0.62	26.23 ± 0.05
26	37	CHN	4	NV	0.08 ± 0.80	31.38 ± 1.01	71.15 ± 4.43	1.14 ± 0.41	31.73 ± 1.00	0.50 ± 0.17	1.11 ± 0.16	29.90 ± 1.07	29.38 ± 0.27
27	47	DEL	3	NV	−0.40 ± 0.83	23.11 ± 0.99	62.59 ± 5.49	0.07 ± 0.05	23.13 ± 0.93	0.80 ± 0.33	1.22 ± 0.36	19.32 ± 1.15	18.71 ± 1.03
28	14	DEL	7	NV	0.64 ± 1.39	31.49 ± 1.53	68.60 ± 5.11	0.62 ± 0.26	31.84 ± 1.48	0.46 ± 0.14	1.16 ± 0.24	30.37 ± 0.32	32.14 ± 0.63
29	31	DEL	12	NV	−0.87 ± 1.02	18.43 ± 2.14	64.32 ± 6.75	0.03 ± 0.04	18.49 ± 2.04	0.96 ± 0.30	1.22 ± 0.29	13.40 ± 2.53	15.83 ± 1.20
30	8	DEL	6	NV	1.00 ± 1.41	32.75 ± 1.37	60.43 ± 6.27	0.74 ± 0.38	33.03 ± 1.37	0.45 ± 0.13	1.13 ± 0.23	35.70 ± 0.00	32.39 ± 0.00
31	14	DEL	8	NV	0.36 ± 1.01	29.19 ± 0.81	73.44 ± 6.14	0.58 ± 0.56	29.44 ± 0.91	0.46 ± 0.08	1.19 ± 0.25	29.43 ± 0.82	30.59 ± 0.36
32	3	DEL	1	NV	−2.33 ± 1.15	14.47 ± 1.33	78.07 ± 4.92	0.00 ± 0.01	15.44 ± 0.79	1.24 ± 0.23	1.33 ± 0.33	12.33 ± 0.00	13.04 ± 0.00
33	28	DEL	11	NV	−0.75 ± 1.11	22.55 ± 2.90	49.29 ± 7.38	0.03 ± 0.05	22.64 ± 2.68	0.82 ± 0.36	1.28 ± 0.28	18.00 ± 2.56	21.98 ± 2.93
34	23	DEL	9	NV	0.43 ± 0.79	31.28 ± 1.52	63.57 ± 5.41	0.63 ± 0.49	31.66 ± 1.45	0.41 ± 0.13	1.09 ± 0.25	32.27 ± 0.21	30.37 ± 0.31
35	2	HYD	6	NV	0.50 ± 0.71	28.90 ± 0.14	67.30 ± 0.42	0.33 ± 0.10	29.21 ± 0.07	0.27 ± 0.01	1.45 ± 0.21	26.92 ± 0.00	31.02 ± 0.00
36	51	HYD	12	NV	−0.45 ± 0.67	25.22 ± 1.13	50.97 ± 5.44	0.13 ± 0.11	25.50 ± 1.17	0.43 ± 0.16	1.30 ± 0.31	21.75 ± 0.55	22.84 ± 0.38
37	25	HYD	3	NV	0.36 ± 0.76	28.60 ± 1.33	54.54 ± 8.53	0.75 ± 0.64	28.87 ± 1.24	0.44 ± 0.25	1.29 ± 0.21	27.27 ± 1.32	25.75 ± 0.70
38	45	HYD	11	NV	−0.69 ± 0.92	26.85 ± 1.44	53.79 ± 9.08	0.29 ± 0.21	27.13 ± 1.41	0.43 ± 0.19	1.26 ± 0.27	23.58 ± 1.37	23.98 ± 0.16
39	39	HYD	10	NV	−0.28 ± 1.00	27.66 ± 1.26	72.91 ± 11.92	0.30 ± 0.21	27.95 ± 1.35	0.45 ± 0.17	1.34 ± 0.31	23.25 ± 1.15	24.93 ± 0.29
40	30	HYD	9	NV	−0.23 ± 1.10	26.96 ± 1.43	79.95 ± 4.29	0.39 ± 0.39	27.34 ± 1.57	0.49 ± 0.28	1.34 ± 0.31	24.50 ± 1.12	25.97 ± 0.05
41	1	HYD	8	NV	0.00	27.60	68.80	0.80	28.05	0.61	1.68	26.00	25.96
42	27	KOL	3	NV	0.04 ± 0.65	28.70 ± 1.08	59.18 ± 5.71	0.67 ± 0.66	28.81 ± 1.13	0.48 ± 0.12	1.16 ± 0.35	24.58 ± 0.96	23.28 ± 0.62
43	75	KOL	6	NV	0.17 ± 0.74	30.83 ± 0.77	79.25 ± 4.97	0.99 ± 0.60	31.13 ± 0.68	0.52 ± 0.21	1.27 ± 0.45	29.56 ± 0.80	29.73 ± 0.19
44	57	KOL	7	NV	0.11 ± 0.72	32.30 ± 0.57	73.99 ± 2.47	1.29 ± 0.51	32.57 ± 0.58	0.50 ± 0.17	1.22 ± 0.47	30.67 ± 1.15	30.19 ± 0.10
45	30	MUM	3	NV	−0.07 ± 0.91	29.68 ± 1.09	44.69 ± 9.58	0.79 ± 0.45	29.94 ± 1.06	0.49 ± 0.12	1.12 ± 0.22	28.55 ± 1.11	27.31 ± 0.07
46	64	SHM	7	NV	−0.06 ± 0.85	23.02 ± 1.82	83.70 ± 5.04	0.05 ± 0.06	23.21 ± 1.89	0.60 ± 0.17	1.25 ± 0.29	24.49 ± 0.75	24.50 ± 0.13

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7.8 (continued)

Bin	Responses	Location	Month	Space Operation Mode	TSV	T _a (°C)	RH (%)	V _a (m/ s)	T _{op} (°C)	CLO	MET	Daily Avg. T _{out} (°C)	T _{out-30DRM} (°C)
47	55	SHM	12	NV	−1.35 ± 0.58	13.28 ± 1.95	52.23 ± 10.22	0.01 ± 0.01	13.45 ± 2.00	1.34 ± 0.36	1.28 ± 0.25	5.65 ± 0.91	8.26 ± 0.28
48	2	SHM	4	NV	0.00 ± 0.00	18.20 ± 0.00	59.15 ± 0.35	0.05 ± 0.05	18.15 ± 0.15	1.14 ± 0.39	1.08 ± 0.00	15.81 ± 0.00	14.43 ± 0.00
49	87	SHM	10	NV	−0.45 ± 0.80	19.40 ± 2.13	44.21 ± 9.33	0.04 ± 0.06	19.57 ± 2.19	0.79 ± 0.25	1.27 ± 0.25	15.03 ± 0.53	18.43 ± 0.36
50	47	SHM	9	NV	0.02 ± 0.68	21.79 ± 1.82	80.95 ± 8.30	0.08 ± 0.07	21.97 ± 1.86	0.63 ± 0.17	1.25 ± 0.35	21.39 ± 1.03	22.91 ± 0.74
51	20	SHM	8	NV	−0.10 ± 0.72	22.29 ± 1.57	85.60 ± 3.46	0.08 ± 0.09	22.70 ± 1.55	0.57 ± 0.17	1.27 ± 0.32	22.94 ± 2.26	24.43 ± 0.21
52	61	SHM	1	NV	−0.84 ± 0.78	13.34 ± 2.42	51.78 ± 9.30	0.02 ± 0.02	13.49 ± 2.49	1.25 ± 0.29	1.28 ± 0.24	7.57 ± 0.24	6.29 ± 0.77
53	15	SHM	5	NV	−0.13 ± 0.99	22.08 ± 2.43	50.53 ± 13.96	0.06 ± 0.10	22.38 ± 2.41	0.55 ± 0.14	1.32 ± 0.23	19.93 ± 2.31	20.00 ± 0.42
54	23	SHM	2	NV	−0.57 ± 0.51	16.36 ± 2.92	66.57 ± 6.61	0.02 ± 0.02	16.68 ± 2.91	0.92 ± 0.20	1.33 ± 0.31	14.05 ± 0.75	9.70 ± 0.59
55	46	SHM	11	NV	−1.63 ± 0.77	14.95 ± 2.01	66.31 ± 5.58	0.03 ± 0.03	15.01 ± 2.05	1.25 ± 0.34	1.27 ± 0.22	8.07 ± 2.67	12.85 ± 1.59
56	12	SHM	6	NV	0.00 ± 0.60	22.54 ± 1.54	80.10 ± 7.49	0.06 ± 0.12	22.77 ± 1.58	0.56 ± 0.10	1.32 ± 0.33	22.97 ± 2.51	21.77 ± 0.96
57	3	SHM	3	NV	−1.00 ± 1.00	12.47 ± 1.00	75.27 ± 4.32	0.00 ± 0.00	12.55 ± 0.89	1.16 ± 0.65	1.18 ± 0.18	10.74 ± 0.00	10.30 ± 0.00
58	6	AMD	4	AC	0.00 ± 0.63	29.12 ± 1.45	82.60 ± 2.91	0.37 ± 0.06	29.49 ± 1.48	0.37 ± 0.08	1.14 ± 0.11	32.76 ± 0.26	30.87 ± 0.36
59	12	AMD	9	AC	−0.25 ± 0.75	27.96 ± 2.41	81.05 ± 2.87	1.24 ± 1.31	28.91 ± 2.20	0.49 ± 0.14	1.23 ± 0.24	30.31 ± 0.34	29.47 ± 0.66
60	9	AMD	10	AC	0.00 ± 0.50	27.53 ± 1.61	81.47 ± 1.44	0.28 ± 0.22	28.01 ± 1.72	0.47 ± 0.14	1.36 ± 0.28	29.36 ± 0.85	30.69 ± 0.06
61	6	AMD	7	AC	−0.17 ± 0.98	23.37 ± 2.24	64.45 ± 1.80	0.45 ± 0.35	24.91 ± 1.83	0.47 ± 0.14	1.11 ± 0.08	31.53 ± 0.00	30.93 ± 0.00
62	15	AMD	6	AC	0.00 ± 0.76	28.60 ± 1.04	57.06 ± 7.45	1.01 ± 0.74	29.20 ± 1.24	0.47 ± 0.10	1.11 ± 0.18	32.13 ± 1.32	32.17 ± 0.18
63	3	CHN	12	AC	−1.67 ± 0.58	22.70 ± 1.04	57.90 ± 1.05	0.39 ± 0.26	23.08 ± 0.33	0.60 ± 0.19	1.03 ± 0.06	27.37 ± 0.00	26.36 ± 0.00
64	1	CHN	8	AC	0.00	29.80	70.70	0.82	29.80	0.91	1.15	29.33	29.64
65	3	CHN	9	AC	−1.00 ± 1.00	27.53 ± 1.31	65.60 ± 14.46	0.96 ± 0.55	27.12 ± 1.12	0.44 ± 0.07	1.20 ± 0.53	28.92 ± 0.38	28.99 ± 0.04
66	9	CHN	2	AC	−1.44 ± 0.73	24.92 ± 1.29	52.12 ± 3.74	0.63 ± 0.28	25.29 ± 1.41	0.54 ± 0.16	1.02 ± 0.13	26.53 ± 0.16	25.87 ± 0.02
67	11	CHN	3	AC	−1.82 ± 0.60	24.16 ± 1.99	58.35 ± 7.08	0.50 ± 0.28	24.64 ± 1.79	0.48 ± 0.16	1.13 ± 0.21	28.80 ± 1.40	27.10 ± 0.86
68	4	CHN	11	AC	−1.50 ± 0.58	24.38 ± 2.88	54.20 ± 10.96	0.66 ± 0.55	24.26 ± 2.51	0.58 ± 0.14	1.18 ± 0.22	27.90 ± 0.25	28.27 ± 0.09
69	5	CHN	10	AC	−1.80 ± 0.45	25.74 ± 1.29	54.08 ± 8.67	0.90 ± 0.78	26.16 ± 1.14	0.45 ± 0.17	1.10 ± 0.06	28.11 ± 0.59	28.93 ± 0.03
70	1	CHN	1	AC	−2.00	24.90	69.50	0.73	24.99	0.82	1.00	27.00	26.26
71	5	CHN	4	AC	−1.60 ± 0.89	28.24 ± 2.15	52.98 ± 11.04	0.66 ± 0.32	28.26 ± 1.93	0.63 ± 0.12	1.16 ± 0.25	30.63 ± 0.36	29.76 ± 0.36
72	8	DEL	7	AC	0.50 ± 0.53	28.43 ± 2.35	62.74 ± 6.49	0.77 ± 0.32	28.42 ± 2.60	0.44 ± 0.16	1.25 ± 0.31	32.03 ± 1.31	31.91 ± 0.72
73	2	DEL	12	AC	−0.50 ± 2.12	20.10 ± 0.42	66.90 ± 0.99	0.09 ± 0.09	20.60 ± 1.64	0.89 ± 0.01	1.45 ± 0.49	15.30 ± 4.18	17.42 ± 0.46
74	6	DEL	6	AC	0.33 ± 1.21	31.47 ± 2.76	58.67 ± 2.64	0.73 ± 0.28	31.80 ± 2.68	0.50 ± 0.11	1.18 ± 0.16	35.70 ± 0.00	32.39 ± 0.00
75	15	DEL	8	AC	−0.13 ± 1.36	27.81 ± 1.54	59.59 ± 7.40	0.40 ± 0.21	28.06 ± 1.43	0.35 ± 0.11	1.20 ± 0.25	29.87 ± 0.90	30.78 ± 0.39
76	2	DEL	9	AC	−0.50 ± 2.12	28.05 ± 1.34	54.50 ± 17.39	0.14 ± 0.13	27.65 ± 0.22	0.45 ± 0.11	0.91 ± 0.05	32.06 ± 0.00	30.05 ± 0.00
77	3	HYD	3	AC	0.00 ± 0.00	25.83 ± 0.85	53.47 ± 0.90	0.33 ± 0.11	26.18 ± 0.46	0.33 ± 0.12	1.12 ± 0.12	26.50 ± 0.00	25.64 ± 0.00
78	14	KOL	7	AC	0.14 ± 0.36	26.71 ± 1.89	55.16 ± 4.67	0.49 ± 0.45	26.95 ± 1.44	0.55 ± 0.11	0.98 ± 0.26	31.61 ± 1.05	30.17 ± 0.09
79	8	SHM	12	AC	−1.13 ± 0.64	12.61 ± 1.51	55.10 ± 4.12	0.04 ± 0.07	13.05 ± 1.74	1.21 ± 0.43	1.38 ± 0.15	5.72 ± 0.98	8.13 ± 0.48
80	9	SHM	1	AC	−1.00 ± 0.87	14.46 ± 1.74	47.50 ± 4.81	0.06 ± 0.10	15.17 ± 1.81	1.25 ± 0.14	1.33 ± 0.16	7.28 ± 0.28	6.16 ± 0.11
81	7	SHM	2	AC	−0.71 ± 0.76	16.17 ± 1.54	64.71 ± 7.42	0.06 ± 0.08	16.80 ± 1.11	1.02 ± 0.20	1.14 ± 0.23	13.58 ± 0.02	9.62 ± 0.39
82	10	SHM	11	AC	−1.60 ± 1.58	14.49 ± 2.04	64.97 ± 4.50	0.03 ± 0.03	15.06 ± 2.22	1.39 ± 0.35	1.10 ± 0.23	6.98 ± 1.84	11.60 ± 0.31

7.9 Bin Significance Checks for Acceptability Band Calculation

Three checks were performed to test the statistical significance of the 82 'Location + Month + Mode' bins; as a result, 11 bins were found to be statistically significant and were used for calculating the 80% and 90% thermal acceptability bands. The three checks were:

1. The bin size (no. of responses) should be at least 20,
2. The p-value of the bin should at most 0.01, and
3. The calculated neutral temperature (T_{neut}) should be within a 1 °C bracket of the indoor air temperature (T_{air}) range.

7.10

Weighted Average Sensitivity Slope Calculation for Significant Bins

Bin ID	Responses (Residence Mode)	Slope	Intercept	R ²	p-value	Responses × Slope
3	81 (NV: 10, MM: 71)	0.135	−3.7	0.61	0.004	10.9
17	69 (NV: 5, MM: 64)	0.424	−12.3	0.72	0.003	29.3
21	45 (NV: 3, MM: 42)	0.246	−7.1	0.65	0.005	11.1
22	51 (NV: 4, MM: 47)	0.217	−6.5	0.83	0.001	11.1
24	77 (NV: 6, MM: 71)	0.327	−10.2	0.71	0	25.2
26	37 (NV: 0, MM: 37)	0.676	−21.4	0.88	0	25.0
36	51 (NV: 18, MM: 33)	0.235	−6.4	0.68	0.003	12.0
47	55 (NV: 46, MM: 9)	0.09	−2.7	0.50	0	5.0
49	87 (NV: 74, MM: 13)	0.166	−3.7	0.72	0	14.4
52	61 (NV: 46, MM: 15)	0.08	−1.9	0.16	0.002	4.9
55	46 (NV: 36, MM: 10)	0.123	−3.6	0.18	0.006	5.7
Weighted Average Sensitivity Slope (Sum of Products/Sum of Responses)						0.23

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